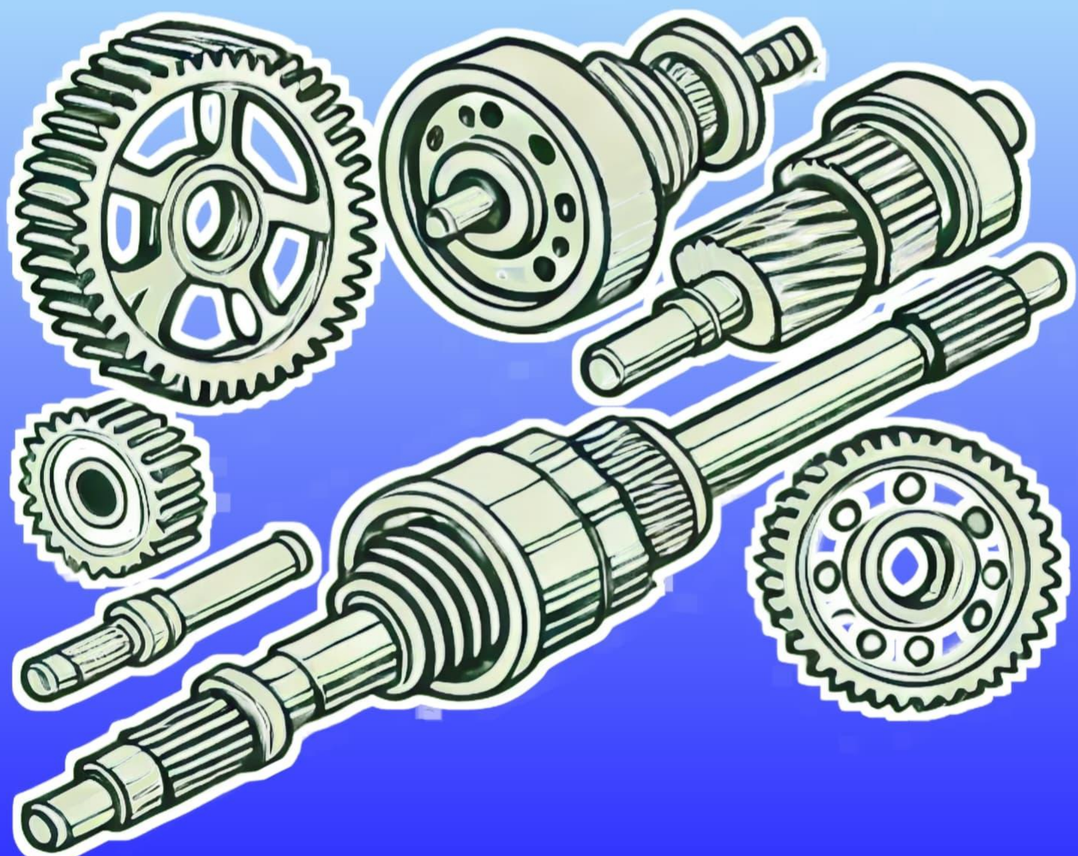
Engineering Mechanics Suggestion (2024-2025)

PDF

For 2nd Semester of Polytechnic

Top 237 Questions with Step by Step solution From All Chapters



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Contents

1. **A force can be represented by-**
   (a) direction
   (b) magnitude
   (c) direction and magnitude
   (d) any one of the above
   Explanation: A force is a vector quantity, meaning it has both magnitude (strength of the force) and direction (the line along which the force acts).
   Answer: (c)

2. **Action and reaction-**
   (a) Always act on same body
   (b) Never act on the same body
   (c) Action act but reaction never act
   (d) None of the above
   Explanation: Action and reaction forces are equal in magnitude but opposite in direction. They are simultaneous in nature as they are applied at the same instant but on different bodies.
   Answer: (b)

3. **When two forces of magnitude 3N and 4N acts at right angles to each other their resultant is a) 14N b) 10N c) 2N d) 5N.**
   Explanation: As the angle between the forces are 90°
   $\therefore$ Resultant $= \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5\text{N}$
   Answer: (d)

4. **The forces which meet at one point but their lines of action do not lie on the same plan, are called**
   (a) Coplanar concurrent forces
   (b) Coplanar non-concurrent forces
   (c) Non-coplanar concurrent forces
   (d) Intersecting forces
   Explanation: As the forces meet at one point. Thus, these forces can be called concurrent. The lines of actions of these forces do not lie in one plane, thus, these forces can be called non-coplanar. Therefore the forces, which meet at one point but their lines of action do not lie in one plane, are called non-coplanar concurrent forces.
   Answer: (c)

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5. **If the tension in the cable supporting an elevator is equal to the weight of the elevator, the elevator may be**
   (a) Going up with increasing speed
   (b) Going down with increasing speed
   (c) Going up with uniform speed
   (d) Going down with decreasing speed
   Explanation: If the tension in the cable of the elevator is the same as the weight of the elevator, it means that the net force on the system is zero. Therefore, the net acceleration of the system will also be zero. This means that the elevator is moving up or down with uniform speed.
   Answer: (c)

6. **Forces are called concurrent when their lines of action meet in**
   (a) one point
   (b) two points
   (c) perpendicular planes
   (d) different planes
   Explanation: Concurrent forces refer to those forces which intersect through a common point.
   Answer: (a)

7. **A force acting on a body may**
   (a) balance the other forces acting on it
   (b) retard its motion
   (c) change its motion
   (d) all of the above
   Explanation: Force can make a body that is at rest to move. It can stop a moving body or slow it down. It can accelerate the speed of a moving body. It can also change the direction of a moving body along with its shape and size.
   Answer: (d)

8. **Which of the following do not have identical dimensions?**
   (a) Momentum and impulse
   (b) Torque and energy
   (c) Torque and work
   (d) Moment of a force and angular momentum
   Explanation: Dimension of moment is $[ML^2T^{-2}]$ and dimension of angular momentum is $[ML^2T^{-1}]$. Therefore their dimensions are not identical
   Answer: (d)

9. Which of the following is not the unit of power?
(a) kW (kiloWatt)
(b) HP (Horse Power)
(c) kcal/sec
(d) Watt.sec

Explanation: Watt.sec is not the unit of power because Watt.sec. $= \frac{\text{Joule}}{\text{Sec.}} \times$ Sec. $=$ Joule. It is SI unit of work or energy.

Answer: (d)

10. The line of action of resultant of two unlike parallel forces lies-
(a) in between them nearer to the smaller force
(b) in between them nearer to the larger force
(c) in the line of action of larger force
(d) no one is correct

Explanation: Let the angle between larger force and resultant is θ,

$\therefore \tan\theta = \frac{Q \sin\alpha}{P + Q\cos\alpha}$

[Here $\alpha = 180°$, for unlike parallel forces]

$\therefore \tan\theta = \frac{Q \sin 180°}{P + Q\cos 180°} = 0$

$\Rightarrow \theta = 0$

So, the resultant lies in the line of action of larger force.

Answer: (c)

11. Define collinear forces.

Collinear Forces: If the lines of action of the forces acting on the body lie on a same line, then the forces are called as collinear forces.

12. Define coplanar forces.

Answer:

Coplanar Forces: If the lines of action of the forces acting on the body lie on the same plane, then the forces are called as Coplanar forces.

13. Define concurrent forces.

Answer:

Concurrent Forces: If the lines of action of the forces acting on the body passes through a common point, then the forces are called as concurrent forces.

14. What are coplanar concurrent forces?

Answer:

Coplanar concurrent forces: The forces which meet at one point and their lines of action lie on the same plane are called coplanar concurrent forces.

15. Define Scalar quantity and vector quantity. Give two examples of each.

Answer:

Scalar Quantity: The physical quantities that have the only magnitude and no direction are known as scalar quantity or scalar. Examples of scalars are electric charge, density, mass etc.

Vector Quantity: The physical quantities that have both magnitudes as well as direction are known as vector quantity or vector. The examples of vectors are velocity, acceleration, force etc.

16. What is composition of forces?

Answer:

Composition of Forces: The process of finding out the resultant force of two or more given forces is called composition of forces.

17. What is Resolution of forces?

Answer:

Resolution of forces: Breaking down a single force into two or more force is known as the resolution of forces.

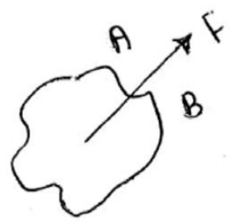
Or, The process of splitting the force into components is called resolution of a force.

The resolution of forces helps in determining the resultant of a number of forces acting on a body as it reduces vector addition to algebraic addition.

18. Explain Bow's notation.

Ans:

Bow's notation: Bow's notation is used designate a force as per this notation, each force is designated or named by two spaces one on each side of the line of action of a force. This space are generally named by capital letter's as A, B, C…… serially.

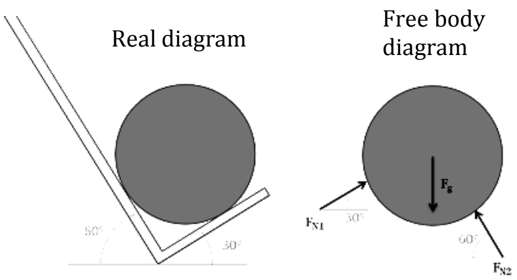


Explanation: A force say 'F' acting on rigid body divided space above or below it into two parts, say A and B hence the force 'F' is named as AB.

19. What is free body diagram?

Answer:

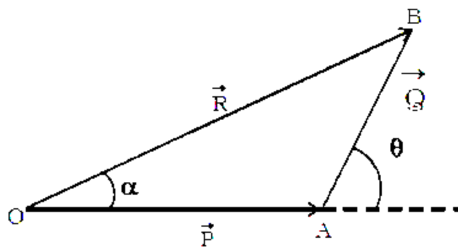
Free Body Diagram: A free Body Diagram is a graphical representation of the applied forces, moments, and consequent reactions on a body in a particular situation.



20. State triangle law of forces.

Answer:

Triangle law of force: If two forces are simultaneously acting on a particle and are represented in terms of magnitude and direction by two sides of a triangle, taken in an order, then the third side or the closing side of triangle, drawn in opposite order, represents the resultant of above two forces in terms of magnitude and direction.



Let us assume two forces $\vec{P}$ and $\vec{Q}$ represented in terms of magnitude and direction by OA and AB of triangle ABC . Then OB represents resultant of P and Q

$\therefore \vec{R} = \vec{P} + \vec{Q}$

And $R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$

$\tan\theta = \frac{Q\sin\alpha}{P + Q\cos\alpha}$

21. When two forces of magnitude 3N and 4N acts at right angles to each other. Find the magnitude of their resultant.

Answer:

We know, if two forces P and Q acts at angle of α and if their resultant be R then,

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$$

Here, $P = 3N,\ Q = 4N$ and $\alpha = 90°$

$\therefore R = \sqrt{3^2 + 4^2 + 2.3.4\cos 90°}$

$= \sqrt{9 + 16 + 0} = \sqrt{25} = 5N$ (Ans)

22. If two equal forces each of magnitude P act at an angle θ, then what will be their resultant?

Answer:

We know, if two forces P and Q acts at angle of α and if their resultant be R then,

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$$

Here, $P = Q$ and $\alpha = \theta$

So, resultant $R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$

$\Rightarrow R = \sqrt{P^2 + P^2 + 2P.P\cos\theta}$

$= \sqrt{2P^2 + 2P^2\cos\theta}$

$= \sqrt{2P^2(1 + \cos\theta)}$

$= P\sqrt{2.2\cos^2\frac{\theta}{2}} = 2P\cos\frac{\theta}{2}$ (Ans)

23. What is rigid body?

Answer:

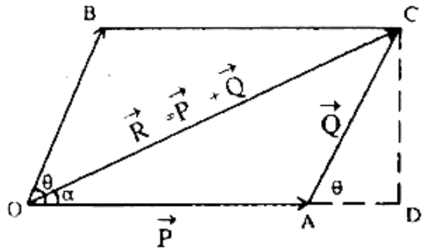
Rigid Body: A Rigid Body is generally defined as a body on which the distance between two points never changes whatever be the force applied on it. Or in other words, it can be said that the body which does not deform under the influence of forces is known as a Rigid Body.

As in real life, no substance is rigid. One general real-time example of a Rigid Body is a ball bearing made of hardened steel is a good example of a Rigid Body.

24. Sate Parallelogram law of forces. Draw sketch to explain.

Answer:

Parallelogram law of forces: If two forces are acting simultaneously on a particle and are represented in terms of magnitude and direction of two adjacent sides of a parallelogram, then the diagonal passing through point of intersection of these forces represents the resultant of above two forces.

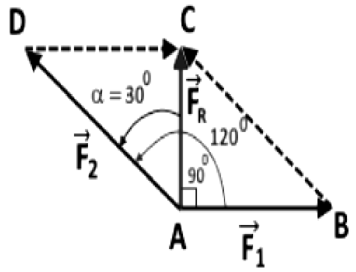


Suppose, two forces, $\vec{P}$ and $\vec{Q}$ acting at point O be represented by two sides of a parallelogram, then their resultant $\vec{R}$ is represented by diagonal OC of the parallelogram $OACB$.

25. Two forces act an angle of 120°. The bigger force is of 40N and the resultant is perpendicular to the smaller one, find the smaller force.

Solution:

Let the resultant $\overrightarrow{F_R}$ is making an angle 90° with the smaller force $\overrightarrow{F_1}$. This implies that the resultant makes an angle α=120–90=30° with the bigger vector $\overrightarrow{F_2}$ as shown in the figure.



From the right angle triangle ACD,
defining $\sin\alpha$ we get

$\sin\alpha = \frac{DC}{AD}$

$AD = |\overrightarrow{F_2}| = 40N$ (given)

$DC = |\overrightarrow{F_1}| = ?$

$\sin 30° = \frac{|\overrightarrow{F_1}|}{|\overrightarrow{F_2}|}$

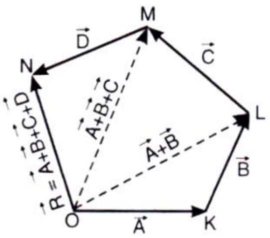
$\Rightarrow |\overrightarrow{F_1}| = |\overrightarrow{F_2}| \times \sin 30°$

$= 40 \times \frac{1}{2} = 20N$ (Ans)

26. State the polygon law of force.

Answer:

Polygon Law of Force: If any number of forces acting at a point can be represented in direction and magnitude by the sides of a polygon taken in order, their resultant may be represented in magnitude and direction by the closing side of the polygon, taken in the opposite order.



27. Explain how to subtract two vectors.

Answer:

Vector Subtraction: The subtraction of two vectors is similar to addition. Suppose $\vec{P}$ is to be subtracted from $\vec{Q}$. $(\vec{P} - \vec{Q})$ can be said as the addition of $\vec{P}$ and $(-\vec{Q})$. Thus, the addition formula can be applied as:

$$\vec{P} - \vec{Q} = \sqrt{P^2 + Q^2 - 2PQ\cos\theta}$$

28. The following forces act at a point –

i) $20N$ inclined at $30°$ to north of east.

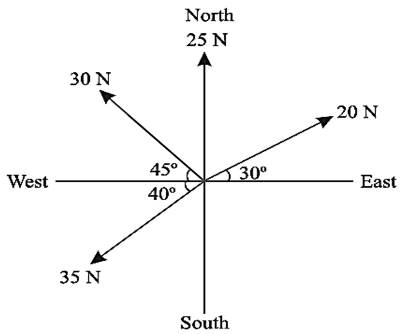
ii) $25N$ towards north

iii) $30N$ towards north-west

iv) $35N$ inclined at $40°$ to south of west.

Find the magnitude and direction of the resultant of the force system.

Solution:



Resolving all the forces horizontally i.e. along east-west we get,

$\sum H = 20\cos 30° - 30\cos 45° - 35\cos 40°$
$= -30.7N$

And resolving all the forces vertically i.e. along north-south line we get,

$\sum V = 25 + 20\sin 30° + 30\sin 45° - 35\sin 40°$
$= 33.7N$

∴ Magnitude of resultant

$= \sqrt{(\sum H)^2 + (\sum V)^2}$

$= \sqrt{(-30.7)^2 + (33.7)^2} = 45.6N$ (Ans)

Let, direction of resultant is θ with respect to east.

∵ $\sum H$ is negative and $\sum V$ is positive.

∴ θ lies between 90° to 180°.

$\therefore \theta = 180° - \tan^{-1}\left(\frac{33.7}{30.7}\right)$

$= 180° - \tan^{-1}(0.911)$

$= 180° - 47.67° = 132.33°$ (Ans)

29. If two equal forces acting simultaneously at a point and if their resultant is equal to any of them, then find the angle between the forces.

Solution:

We know, if two forces P and Q acts at angle of α and if their resultant be R then,

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$$

Here, $P = Q = R$

$\therefore P = \sqrt{P^2 + P^2 + 2P.P\cos\alpha}$

$\Rightarrow P^2 = 2P^2 + 2P^2\cos\alpha$

$\Rightarrow P^2 = 2P^2(1 + \cos\alpha)$

$\Rightarrow 1 + \cos\alpha = \frac{P^2}{2P^2} = \frac{1}{2}$

$\Rightarrow \cos\alpha = \frac{1}{2} - 1 = -\frac{1}{2} = \cos 120°$

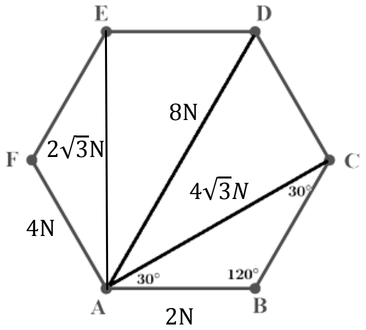
$\Rightarrow \alpha = 120°$ (Ans)

30. $ABCDEF$ is a regular hexagon. Forces of magnitude $2N, 4\sqrt{3}N, 8N, 2\sqrt{3}N$, and $4N$ act at A in direction of AB, AC, AD, AE and AF respectively determine the resultant completely.

Solution:

Given, $\overrightarrow{AB} = 2N, \overrightarrow{AC} = 4\sqrt{3}N, \overrightarrow{AD} = 8N$

$\overrightarrow{AE} = 2\sqrt{3}N,\ \overrightarrow{AF} = 4N$



The lines AE and AB are perpendicular to each other and are taken as the Y - axis and X- axis respectively .The vectors are resolved along the X - and the Y – axis we get,

$\sum F_x = 2 + 4\sqrt{3}\cos 30° + 8\cos 60° - 4\sin 30°$

$\quad = 10N$

$\sum F_y = 4\sqrt{3}\sin 30° + 8\sin 60° + 2\sqrt{3} + 4\cos 30°$

$\quad = 10\sqrt{3}N$

∴ Magnitude of resultant

$= \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$

$= \sqrt{10^2 + (10\sqrt{3})^2} = 20N$ (Ans)

∴ Direction of resultant with respect to +ve X-axis,

$\theta = \tan^{-1}\left(\frac{10\sqrt{3}}{10}\right)$

$\quad = \tan^{-1}\sqrt{3} = 60°$ (Ans)

31. Two forces of magnitude '2P' and 'P' act on a particle. If the first force be doubled and the second force is increased by 12kN, the direction of their resultant remains unaltered. Find the value of P.

Solution: Let, the angle between given two forces is θ and resultant make angle α with '2P' force.

∴ In first case: $\tan\alpha = \frac{P\sin\theta}{2P + P\cos\theta}$

In second case: $\tan\alpha = \frac{(P+12)\sin\theta}{4P + (P+12)\cos\theta}$

$\therefore \frac{P\sin\theta}{2P + P\cos\theta} = \frac{(P+12)\sin\theta}{4P + (P+12)\cos\theta}$

$\Rightarrow \frac{P\sin\theta}{P(2+\cos\theta)} = \frac{(P+12)\sin\theta}{4P + (P+12)\cos\theta}$

$\Rightarrow \frac{1}{2+\cos\theta} = \frac{(P+12)}{4P + (P+12)\cos\theta}$

$\Rightarrow 4P + (P+12)\cos\theta$

$\quad = 2(P+12) + (P+12)\cos\theta$

$\Rightarrow 4P = 2P + 24, \Rightarrow 2P = 24, \Rightarrow P = 12$ kN (Ans)

32. The resultant of two forces P and Q is R. If Q is doubled, R is doubled and Q is reversed, R is again doubled. Show that $P : Q : R = \sqrt{2} : \sqrt{3} : \sqrt{2}$.

Answer:

Let take the angle between P and Q is θ.

The force expression according to given conditions are given as,

$P^2 + Q^2 + 2PQ\cos\theta = R^2$ ...... (1)

When Q is doubled,

$P^2 + 4Q^2 + 4PQ\cos\theta = 4R^2$ ...... (2)

When Q is reversed,

$P^2 + Q^2 - 2PQ\cos\theta = 4R^2$ (3)

Doing (1)+(3)

$P^2 + Q^2 + 2PQ\cos\theta = R^2$

$P^2 + Q^2 - 2PQ\cos\theta = 4R^2$

$2P^2 + 2Q^2 = 5R^2$ .... (4)

Doing (2)+2×(3)

$P^2 + 4Q^2 + 4PQ\cos\theta = 4R^2$

$2P^2 + 2Q^2 - 4PQ\cos\theta = 8R^2$

$3P^2 + 6Q^2 = 12R^2$

$\Rightarrow P^2 + 2Q^2 = 4R^2$ .... (5)

Doing (4)−(5)

$2P^2 + 2Q^2 = 5R^2$

$-P^2 - 2Q^2 = -4R^2$

$P^2 = R^2$

Dividing eq^n (4) by (5)

$\frac{2P^2+2Q^2}{P^2+2Q^2} = \frac{5R^2}{4R^2} = \frac{5}{4}$

$\Rightarrow 8P^2 + 8Q^2 = 5P^2 + 10Q^2$

$\Rightarrow 3P^2 = 2Q^2, \Rightarrow \frac{P^2}{2} = \frac{Q^2}{3}$

$\therefore \frac{P^2}{2} = \frac{Q^2}{3} = \frac{P^2}{2}$ $[\because P^2 = R^2]$

$\Rightarrow P^2 : Q^2 : R^2 = 2 : 3 : 2$

$\Rightarrow P : Q : R = \sqrt{2} : \sqrt{3} : \sqrt{2}$ (Proved)

33. Compare rigid body and flexible body.

Answer:

Rigid body versus flexible body: Rigid bodies do not deform (stretch, compress, or bend) when subjected to loads, while flexible bodies or deformable bodies are being significantly stretched, compressed, or bent during the period of analysis.

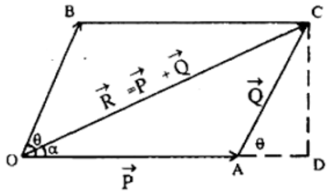
In actuality, no physical body is completely rigid, but most bodies deform so little that this deformation has a minimal impact on the analysis.

{Prepared by NatiTute YouTube Channel}

34. Two concurrent forces P and Q are acting at an angle θ, derive the expression for their resultant force.

Answer:

P and Q are two forces acting at a point O, making an angle θ. They are represented by OA and OB. The diagonal OC of the parallelogram $OBCA$ represents the resultant R.



Draw CD perpendicular to an extended line of OA. In the right angle triangle ADC

$\frac{AD}{AC} = \cos\theta, \Rightarrow AD = AC\cos\theta$

$\frac{CD}{AC} = \sin\theta, \Rightarrow CD = AC\cos\theta$

In the right angle triangle ODC.

$OC^2 = OD^2 + CD^2$

$= (OA + AD)^2 + CD^2$

$= OA^2 + 2 \times OA.AD + AD^2 + CD^2$

$= OA^2 + 2\, OA.AC\cos\theta + AC^2$

$\Rightarrow R^2 = P^2 + 2PQ\cos\theta + Q^2$

∴ Magnitude of resultant

$$R = \sqrt{P^2 + Q^2 + 2PQ \cdot \cos\theta}$$

To find the direction:

If α is the angle made by R with P, then from the right angled triangle ODC,

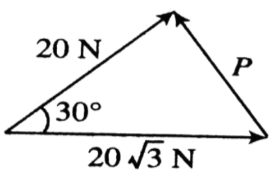
$\tan\alpha = \frac{CD}{OD} = \frac{CD}{OA+AD} = \frac{AC\sin\theta}{OA+AC\cos\theta}$

$\Rightarrow \tan\alpha = \frac{Q\sin\theta}{P+Q\cos\theta}$

35. The resultant of two forces has magnitude 20N. One of the forces is of magnitude 20√3N and makes an angle of 30∘ with the resultant. Then what is the magnitude of the other force?

Answer:

Let P be unknown force.



$\therefore P^2 = \left(20\sqrt{3}\right)^2 + (20)^2 - 2.20\sqrt{3}.20\cos 30°$

$\Rightarrow P^2 = 1200 + 400 - 800\sqrt{3} \times \frac{\sqrt{3}}{2}$

$\Rightarrow P^2 = 1600 - 1200$

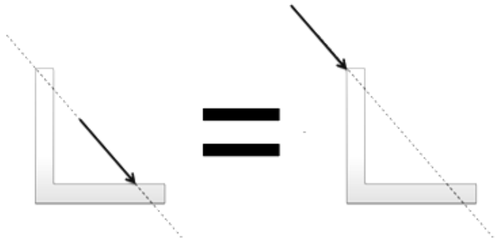
$\Rightarrow P = \sqrt{400} = 20N$ (Ans)

36. State principle of transmissibility of force.

Answer:

Principle of Transmissibility of Force: The point of application of a force can be moved anywhere along its line of action without changing the external reaction forces on a rigid body.

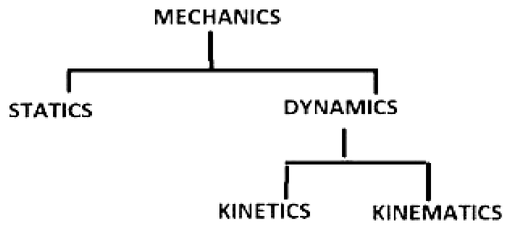
Any force that has the same magnitude and direction, and which has a point of application somewhere along the same line of action will cause the same acceleration and will result in the same moment.



37. Classify branches of mechanics.

Answer:

Mechanics can be classified into the following two main branches: i) Statics ii) Dynamics



Statics: It is a branch of Mechanics that deals with the analysis of loads and their effects on the system when its velocity is zero or it has a static balance with the environment.

Dynamics: It is an important branch of engineering mechanics that deals with forces and their effects when the body is in motion. Dynamics is the study of force when the body is moving.

Kinetics: It is an important branch of dynamics that deals with the movement of bodies due to the use of force.

Kinematics: It is an important branch of dynamics, which deals with bodies in motion without considering the force for movement of the body.

38. What is the maximum and minimum sum of two vectors?

Answer: The maximum sum of two vectors is obtained when the two vectors are directed in the same direction. The minimum sum is obtained when the two vectors are directed in the opposite direction.

1. **The algebraic sum of moments of forces about any point in their plane is equal to the moment of their resultant about the same point- this is the statement according to- a) Varignon's theorem b) Lami's theorem c) coplanar force theorem d) triangular theorem.**
   Explanation: According to Varignon's theorem, "The total of the moments of many coplanar forces around a point equals the moment of the resultant of those forces, or the moment of a force around a point equals the sum of its components." Vaignon's theorem is also known as the "principle of moments."
   Answer: (a)

2. **If the arm of a couple is doubled its moment will be-**
   (a) doubled
   (b) halved
   (c) remain same
   (d) four times
   Explanation: The magnitude of the moment of a couple is given by $|M| = F.L$, where F is the magnitude of either of the forces and L is the arm of the couple. So if L is doubled the moment will be doubled.
   Answer: (a)

3. **A couple is a system of forces with-**
   (a) a resultant force as well as a resultant moment
   (b) neither a resultant force nor a resultant moment
   (c) a resultant moment but no resultant force
   (d) a resultant force but no resultant moment
   Explanation: Since a couple consists of two equal and opposite forces separated by some distance, the resultant force is zero. However it produces a turning effect and its moment is non-zero.
   Answer: (c)

4. **When trying to turn a key into a lock, following is applied**
   (a) coplanar force
   (b) lever
   (c) moment
   (d) couple
   Explanation: A couple is a pair of two equal and opposite forces acting on a body at two different points.
   Answer: (d)

5. **Moment of a force-**
   (a) Varies directly with its distance from the pivot
   (b) Varies inversely with its distance from the pivot
   (c) Is independent of its distance from the pivot
   (d) none of these
   Explanation:
   The magnitude of the moment about pivot:
   $$M = F \times d$$
   Therefore moment is directly proportional to distance from pivot.
   Answer: (a)

6. **A couple consist of**
   (a) Two like parallel forces of same magnitudes
   (b) Two like parallel forces of different magnitudes
   (c) Two unlike parallel forces of same magnitudes
   (d) Two unlike parallel forces of different magnitudes
   Explanation: A couple consists of two parallel forces that are equal in magnitude, opposite in sense and do not share a line of action. It does not produce any translation, only rotation.
   Answer: (c)

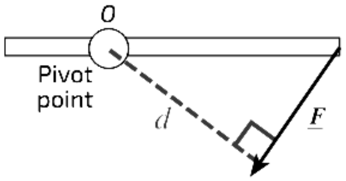
7. **The SI unit of moment is-**
   (a) Nm
   (b) N/m
   (c) Ns
   (d) N/s
   Explanation:
   Moment = Force × Distance
   ∴ Unit of moment = Unit of Force × Unit of distance
   = Newton × Meter
   =Nm
   Answer: (a)

8. **The product of either force of couple with the arm of the couple is called**
   (a) **resultant couple**
   (b) **moment of the forces**
   (c) **resulting couple**
   (d) **moment of the couple**
   Explanation: The moment of a couple is defined as the product of either of the forces of a couple and the perpendicular distance between the line of action of forces.
   Answer: (d)

9. Define moment of force and state its SI unit.

Answer:

Moment of Force (or Moment): The turning effect of force is known as moment of force. It is the product of the force multiplied by the perpendicular distance from the line of action of the force to the pivot or point where the object will turn.



A force of magnitude F acts a perpendicular distance d from the point of rotation O, the magnitude of the moment about that point is:

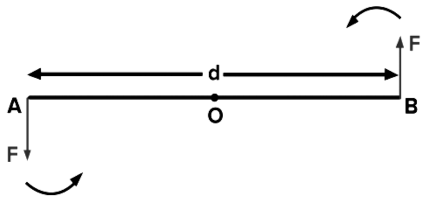
$$M = F \times d$$

The S.I unit is newton metre (Nm).

10. Define moment of a couple. Write its SI unit.

Answer:

Moment of Couple: A couple is a pair of two equal and opposite forces acting on a body at two different points. The moment of a couple is defined as the product of either of the forces of a couple and the perpendicular distance between the line of action of forces (F).



$$Moment of couple = either force \times arm\ of couple$$
$$= F \times d$$

The perpendicular distance between the two forces is called the arm of the couple.

The S.I. unit of moment of a couple is Nm.

11. What is the equivalent force couple?

Answer:

Equivalent force couple: Any set of forces on a body can be replaced by a single force and a single couple acting that is statically equivalent to the original set of forces and moments. This set of an equivalent force and a couple is known as the equivalent force couple system.

12. Define the properties of couple.

Answer:

Properties or Characteristics of couple: The main characteristics of a couple are:

i) The algebraic sum of the forces consisting the couple is zero.

ii) The algebraic sum of the moment of the forces constituting the couple about any point is the same and equal to the moment of the couple itself.

iii) Any number of coplanar couples can be reduced to a single couple.

iv) The couple does not produce translational motion because the two forces that make up the couple are equal and in opposition.

v) Pure rotational motion in the body results from the algebraic sum of the moments of the two forces around any point in their plane is not zero.

13. What is the physical significance of moment?

Answer:

Physical significance of moment:

(a) The moment of a force about some point quantifies its tendency to rotate an object about that point.

(b)The magnitude of moment specifies the magnitude of rotational force.

(c) The direction of a moment specifies the axis of rotation associated with rotational force, following the right hand screw convention.

14. Give some practical examples of an application of couple.

Solution: Following are the examples which utilize couple forces:

(1) Forces exerted by one's hand on a screw-driver.
(2) Forces applied while opening and closing the cap of a bottle.
(3) Forces applied while turning a water tap, cork screw, door key.
(4) Forces applied to the steering of a car.
(5) Forces applied to the pedals of a bicycle.

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15. State and explain Varignon's theorem.

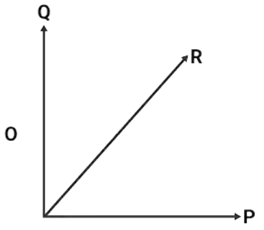
Answer:

Varignon's Theorem: According to Varignon's theorem, "The total of the moments of many coplanar forces around a point equals the moment of the resultant of those forces, or the moment of a force around a point equals the sum of its components."

Vaignon's theorem is also known as the "principle of moments."

Explanation:

Now, look at the figure for the component of forces where R is the resultant of P and Q while P and Q are the forces component of R.



So, if find the moment of R with respect to O. Then, according to Varignon's Theorem,

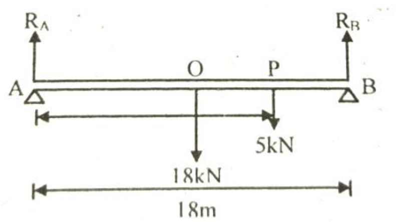
Moment of R About O

= (Moment of P About O + Moment of Q About O)

$$\Rightarrow M_R = M_P + M_Q$$

16. A horizontal bridge (AB) of 18m long weighs 18kN and rest on two supports at its ends. What will be the pressure on the each support when a lorry of weight 5kN starting from A is two third of the way across the bridge.

Solution:



$AP = \frac{2}{3}$ of AB (given) $= \frac{2}{3} \times 18 = 12m$

$AO = \frac{1}{2}$ of $AB = \frac{1}{2} \times 18 = 9m$

For equilibrium we can say sum of force=0

$\therefore R_A + R_B - 18 - 5 = 0$

$\Rightarrow R_A + R_B = 23$ kN

Taking moment about point A it can be written,

$(R_A \times 0) + (R_B \times 18) - (18 \times 9) - (5 \times 12) = 0$

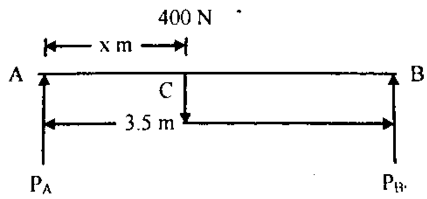
$\Rightarrow 18R_B - 162 - 60 = 0$

$\Rightarrow 18R_B = 222, \Rightarrow R_B = \frac{222}{18} = 12.33$ kN (Ans)

$\therefore R_A = 23 - 12.33 = 10.67$ kN (Ans)

17. Two men carry a concrete block weighing 400N using a light wooden plank on their shoulders. The stronger man can carry 225N. How should the block be supported so that the stronger man gets his share if the length of the plank between the shoulders is 3.5m?

Solution:



Suppose two men are located at point A and B.

Let, A is stronger man and the load placed at x from A.

A can carry 225N so $P_A = 225N$

Again P_A and P_B are the reaction of two men.

$\therefore P_A + P_B = 400N$

$\Rightarrow P_A = 400 - P_B = 400 - 225 = 175N$

Taking moment about A,

$(P_A \times 0) + (P_B \times 3.5) - (400 \times x) = 0$

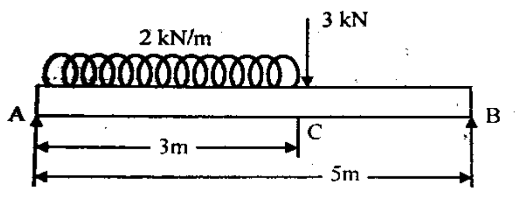
$\Rightarrow 175 \times 3.5 = 400 \times x$

$\Rightarrow x = \frac{175 \times 3.5}{400} = 1.531m$

$\therefore$ The load should be placed at 1.531m from stronger man.

18. A beam AB is simply supported over a span of 5m. It is subjected to uniformly distributed loading 2kN/m over a span of 3m from support A towards B and a concentrated load of 3kN at a distance of 2m from B. Find support reaction at A and B.

Solution:



Total load due to UDL = $2 \times 3 = 6$ kN

This load of 6kN will be acting at the middle point of AC.

Let, R_A and R_B are the reactions at A and B.

$\therefore R_A + R_B = 6 + 3 = 9$kN

Taking moment of all forces about A,

$(R_A \times 0) + (R_B \times 5) - \left(6 \times \frac{3}{2}\right) - (3 \times 3) = 0$

$\Rightarrow 5R_B - 9 - 9 = 0$

$\Rightarrow 5R_B = 18, \Rightarrow R_B = \frac{18}{5} = 3.6$ kN (Ans)

$\therefore R_A = 9 - 3.6 = 5.4$ kN (Ans)

1. **Lami's theorem deals with the equilibrium of a body under the action of three-**
   (a) coplanar forces
   (b) non-coplanar forces
   (c) coplanar non-concurrent forces
   (d) coplanar concurrent forces
   Explanation: Lami's theorem states that if a body is in equilibrium under the action of three coplanar, concurrent, and non-collinear forces, then the magnitudes of the forces are proportional to the sine of the angles opposite to them.
   Answer: (d)

2. **For conditions of equilibrium, Lami's theorem is applicable when the number of forces acting at a point is- a) two b) three c) four d) any number**
   Explanation: Lami's theorem is applicable when three coplanar concurrent and non-collinear forces act at a point and are in equilibrium.
   Answer: (b)

3. **Lami's theorem is applicable to-**
   (a) Coplanar forces
   (b) Concurrent forces
   (c) Coplanar concurrent forces
   (d) Like parallel forces
   Explanation: Lami's Theorem states that "When three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces".
   Answer: (c)

4. **If a body is in equilibrium, we may conclude that-**
   (a) No force is acting on it is zero
   (b) The moment of the forces about any point is zero
   (c) The resultant of all the forces acting on it is zero.
   (d) bothe (b) and (c)

   Explanation: A body is in equilibrium when the vector sum of external forces and moments is zero. i.e. $\sum F = 0$ and $\sum M = 0$.
   Answer: (d)

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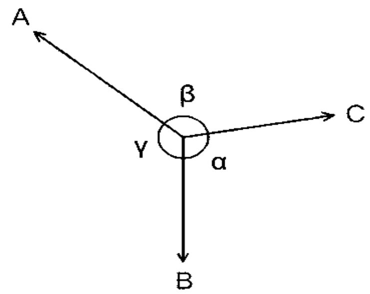
5. **State and explain lami's theorem.**
   Answer:
   Lami's Theorem Statement: Lami's Theorem states that "When three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces".
   Explanation:



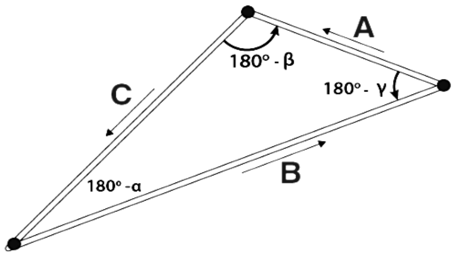
Referring to the above diagram, consider three forces, A, B, and C, acting on a particle or rigid body, making angles α, β and γ with each other.

In the mathematical or equation form, it is expressed as,

$$\frac{A}{sin\alpha} = \frac{B}{sin\beta} = \frac{C}{sin\gamma}$$

6. **Prove Lami's theorem.**
   Answer:
   Prove or Derivation of Lami's Theorem:
   Let F_A, F_B, and F_C be the forces acting at a point. As per the statement of the theorem, we take the sum of all forces acting at a given point which will be zero. i.e., $F_A + F_B + F_C = 0$

   The angles made by force vectors when a triangle is drawn are,



We write angles in terms of complementary angles and use the triangle law of vector addition. Then, by applying the sine rule, we get,

$$\frac{A}{sin\,(180-\alpha)} = \frac{B}{sin\,(180-\beta)} = \frac{C}{sin\,(180-\gamma)}$$

So, we have,

$$\frac{A}{sin\alpha} = \frac{B}{sin\beta} = \frac{C}{sin\gamma} \quad \text{(Proved)}$$

7. Discuss the limitations of Lami's theorem.

Answer:

Limitation of Lami's Theorem:

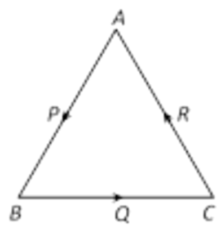
i) There should exist only three forces.
ii) The three forces are to be coplanar
iii) The three forces should remain concurrent.
iv) Those forces should also be non-linear.
v) Forces must have the same nature (inward or outward)
vi) The three angles between those three forces should not be higher than 180 in degrees.

8. State Triangle Law of Equilibrium.

Answer:

Triangle Law of Equilibrium: Three force vectors that form a triangle for which the directions of the forces are all either clockwise around the triangle or counterclockwise around the triangle have a zero resultant, and hence the forces are in equilibrium.

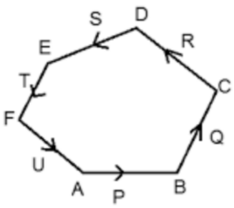
Three forces P, Q and R denoted by three sides of triangle ABC are acting counterclockwise hence the forces are in equilibrium.



9. State Polygon Law of Equilibrium.

Answer:

Polygon Law of Equilibrium: The Polygon Law of Forces states that if a number of concurrent forces acting at a point are represented in magnitude and direction by the sides of a polygon taken in order, then the forces are in equilibrium, and their resultant is zero.

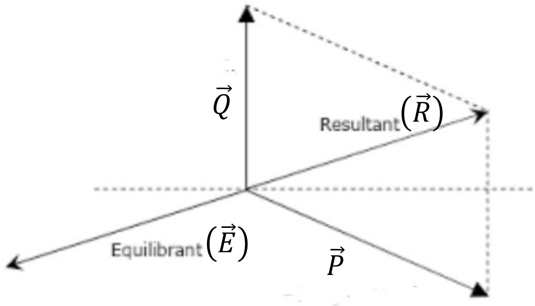


In the above diagram P, Q, R, S, T, and U forces are in equilibrium because they are taken in same order.

10. Define equilibrant force.

Answer:

Equilibrant Force: It is the single force vector needed to establish equilibrium, i.e. to bring about the fact that the resultant of all the forces is zero. It hence has the same magnitude as the original resultant but acts in the opposite direction.



Equilibrant = – Resultant

$\Rightarrow \vec{E} = -\vec{R}$

11. State the conditions of equilibrium.

Answer:

Conditions of Equilibrium: Two conditions of equilibrium of a coplanar force system are-

i) The sum or we can say the resultant of all the external forces that are acting on a body must be equal to zero. Mathematically we can say that,

$\sum_{i=1}^{n} F_i = 0$. We can expand it as,

$$F_1 + F_2 + F_3 + \ldots\ldots\ldots\ldots\ldots F_n = 0$$

ii) The second one says that the sum or resultant of all the external moments from external forces (relative to any point) acting on an object must be equal to zero.

Mathematically we can say that,

$\sum_{i=1}^{n} M_i = 0$. We can expand it as,

$$M_1 + M_2 + M_3 + \ldots\ldots\ldots\ldots\ldots M_n = 0$$

12. What is beam?

Answer:

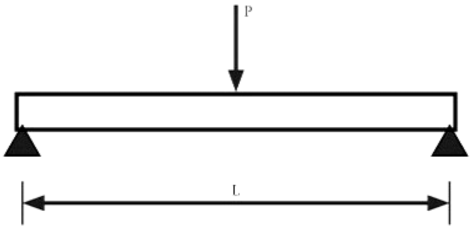
Beam: The beam is defined as the structural horizontal member which is used to bear different loads. It resists the vertical loads, shear forces and bending moments.

13. Describe briefly the types of loading of the beam.

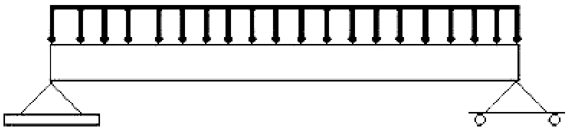
Answer:

Types of Loading of Beam: A Beam may be loaded in a variety of ways. For the analysis purpose it may be splitted in three categories.

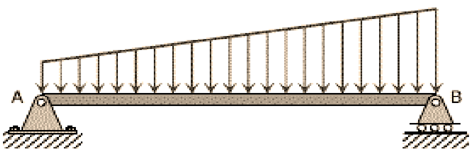
i) **Point or concentrated load:** A concentrated load is the one which acts over a small length that it is assumed to act at a point. Practically, a point load cannot be places as knife edge contact but for calculation purpose we consider that load is being transmitted at a point.

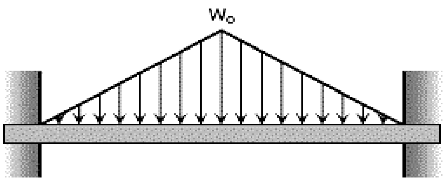


ii) Uniformly Distributed Load (UDL): A distributed load acts over a finite length of the beam. A distributed load may be uniformly. Such loads are measured by their intensity which is expressed by the force per unit distance along the axis of the beam.



iii) Uniformly Varying Load (UVL): A uniformly varying load implies that the intensity of loading increases or decreases at a constant rate along the length.



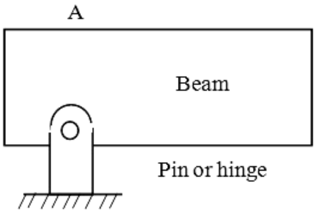


14. Describe briefly the types of support of the beam.

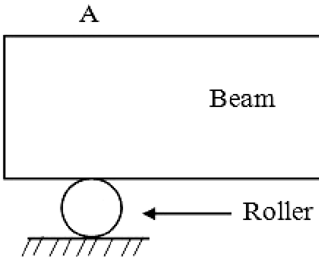
Answer:

Types of Beam Support: The beams usually have three different types of support:

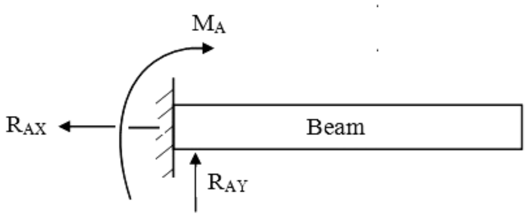
i) Hinged or Pinned Support: The hinged support is capable of resisting force acting in any direction of the plane. Usually, at hinged end the beam is free to rotate but translational displacement is not possible.



ii) Roller Support: The roller support is capable of resisting a force in only one specific line or action. The roller can resist only a vertical force or a force normal to the plane on which roller moves.



iii) Fixed Support: The fixed support is capable of resisting of force in any direction and is also capable of resisting a couple or a moment. A system of three forces can exist at such a support.



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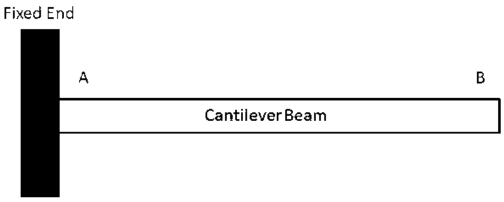
15. Describe briefly the types of beams.

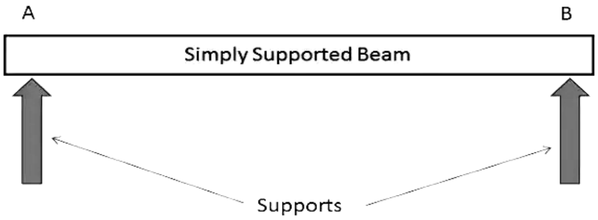
Answer:

Types of Beams: The different types of beams are:

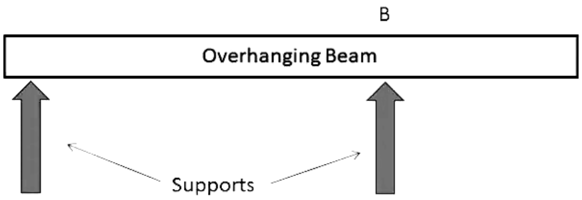
i) Cantilever Beam: A cantilever beam is a beam that is fixed from one end and free at the other end.



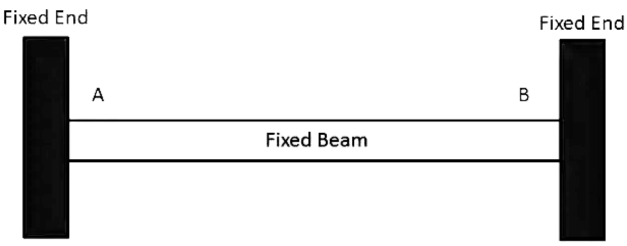
ii) Simply Supported Beam: A beam which is supported or resting on the supports at its both the ends, is called simply supported beam.



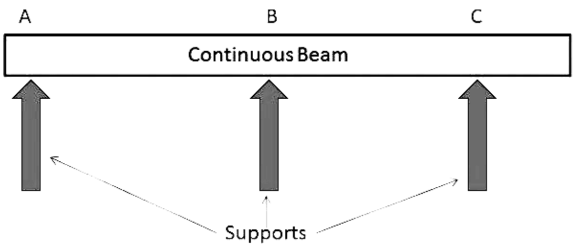
iii) Overhanging Beam: In a beam, if one of its ends is extended beyond the support, it is known as overhanging beam.



iv) Fixed Beams: A beam which has both of its ends fixed or built in walls is called fixed beam.

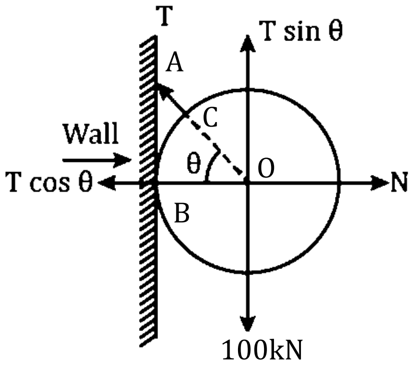


v) Continuous Beam: It is a beam which is provided with more than two supports as shown in figure.



16. A string AC of length 24cm is attached to a point on a smooth vertical wall and to a point on the surface of a sphere of radius CO is 12cm. The sphere whose weight is 100kN hangs in equilibrium against the wall. Find the tension in the string and reaction of the wall.

Solution:



Let, tension in the string is T and reaction of the wall is N and $\angle BOA=\theta$.

Given, CO=BO=12cm, AC=24cm

∴ AO=12+24=36cm

$\therefore \cos\theta = \frac{BO}{AO} = \frac{12}{36} = \frac{1}{3}$

$\Rightarrow \sin\theta = \sqrt{1-\left(\frac{1}{3}\right)^2} = \sqrt{1-\frac{1}{9}} = \frac{2\sqrt{2}}{3}$

As sphere is at equilibrium,

$T\sin\theta = 100\text{ kN}$

$\Rightarrow T \times \frac{2\sqrt{2}}{3} = 100$

$\Rightarrow T = 100 \times \frac{3}{2\sqrt{2}} = 106.06\text{kN}$ (Ans)

And, $T\cos\theta = N$

$\Rightarrow N = 106.06 \times \frac{1}{3} = 35.35\text{kN}$ (Ans)

17. Two forces of 100 N and 150 N are acting simultaneously at a point. What is the resultant of these two forces, if the angle between them is 45°?

Solution:

Given, forces $P = 100\text{N}$, $Q = 150\text{N}$

Angle between P and Q is $\alpha = 45°$

Magnitude of resultant

$R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$

$= \sqrt{100^2 + 150^2 + 2 \times 100 \times 150 \times \cos 45°}$

$= 231.76\text{N}$ (Ans)

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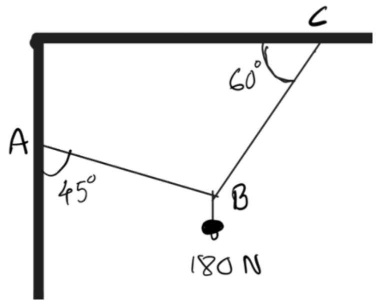
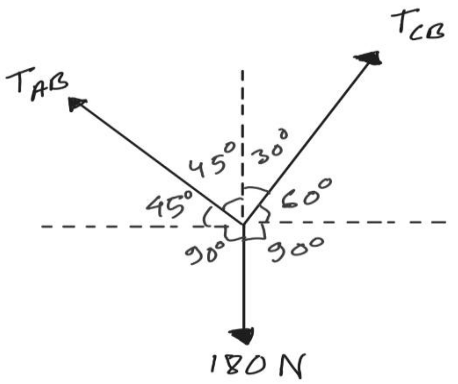
18. 2) Find the Forces (tension) developed in the wire AB & CB, supporting an electric fixture as shown in the adjacent Fig.



Solution: From the geometry of the figure we can get angles as shown in the following figure.



So, angle between forces T_{AB} and T_{CB}

$= 45° + 30° = 75°$

Angle between forces T_{CB} and 180 N

$= 60° + 90° = 150°$

Angle between forces 180 N and T_{AB}

$= 90° + 45° = 135°$

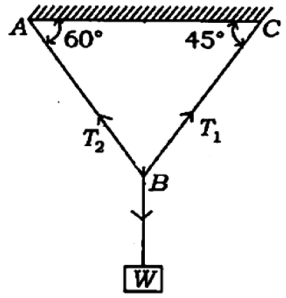
Now, applying Lami's theorem at C we get,

$$\frac{T_{AB}}{\sin 150°} = \frac{T_{CB}}{\sin 135°} = \frac{180}{\sin 75°}$$

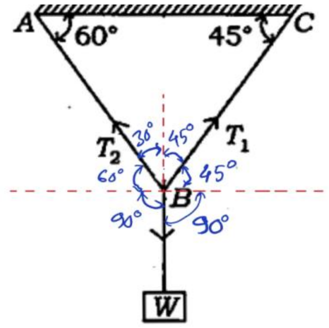
$$\therefore T_{AB} = \frac{180 \times \sin 150°}{\sin 75°} = \frac{180 \times 0.5}{0.97} = 92.78\ N \text{ (Ans)}$$

$$T_{CB} = \frac{180 \times \sin 135°}{\sin 75°} = \frac{180 \times 0.71}{0.97} = 131.75\ N \text{ (Ans)}$$

19. A weight of 2 kN is supported by two strings as shown below. Find the tensions in the strings.



Solution: Given $W = 2kN = 2000N$



Angle between the forces T_1 and W

$= 45° + 90° = 135°$

Angle between the forces W and T_2

$= 90° + 60° = 150°$

Angle between the forces T_1 and T_2

$= 45° + 30° = 75°$

Now, applying Lami's theorem at B we get,

$$\frac{T_1}{\sin 150°} = \frac{T_2}{\sin 135°} = \frac{2000}{\sin 75°}$$

$$\therefore T_1 = \frac{2000 \times \sin 150°}{\sin 75°} = \frac{2000 \times 0.5}{0.97} = 1030.93\ N \text{ (Ans)}$$

$$T_2 = \frac{2000 \times \sin 135°}{\sin 75°} = \frac{2000 \times 0.71}{0.97} = 1463.92\ N \text{ (Ans)}$$

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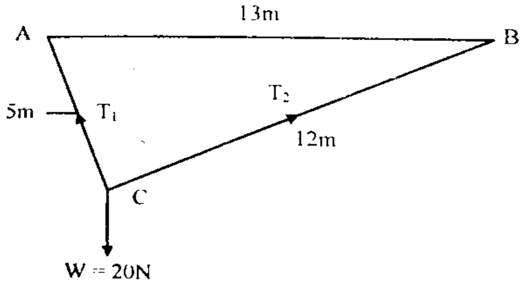
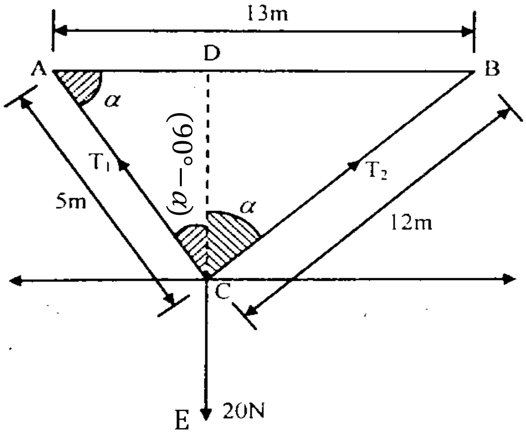
20. A body of weight 20N is suspended by two strings 5m and 12m long and other ends being fastened to the extremities of a rod of length 13m. If the rod is to remain horizontal, determine the tensions in the strings. (Figure below)



Solution:



From the given figure, we get , ∠ACB=90°

As $5^2 + 12^2 = 13^2$

∴ ABC is a right angle triangle.

Let, ∠BCD=α, ∴ $\angle CAD = \alpha, \Rightarrow \sin\alpha = \frac{12}{13}$

$\Rightarrow \cos\alpha = \sqrt{1-\left(\frac{12}{13}\right)^2} = \frac{5}{13}$

Applying Lami's theorem we get,

$\frac{T_1}{\sin \angle BCE} = \frac{T_2}{\sin \angle ACE} = \frac{20}{\sin 90°}$

$\frac{T_1}{\sin(180°-\alpha)} = 20$

$\Rightarrow T_1 = 20\sin\alpha = 20 \times \frac{12}{13} = 18.46\ N$ (Ans)

Again, $\frac{T_2}{\sin \angle ACE} = \frac{20}{\sin 90°}, \Rightarrow \frac{T_2}{\sin(90°+\alpha)} = 20$

$\Rightarrow T_2 = 20\cos\alpha = 20 \times \frac{5}{13} = 7.69\ N$ (Ans)

21. A uniform wheel of 600mm diameter, weighing 5kN rests against a rigid rectangular block of 150mm height. Find the least pull, through the centre of the wheel, required just to turn the wheel over the corner of the block. Also, find the reaction of the block. Consider all the surfaces to be smooth.

Solution:

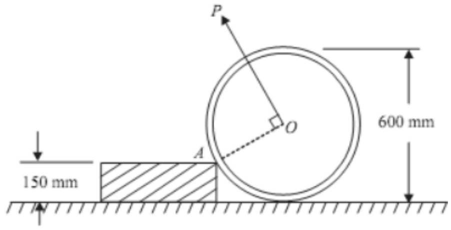
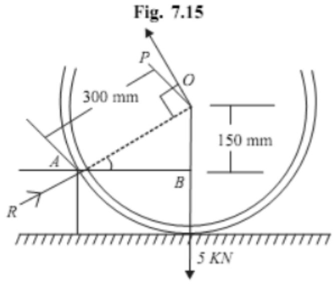


Fig. 7.15



Solution:

Let P = least pull required to turn the wheel.

Least pull should be applied normal to AO.

From the figure $\sin\theta = \frac{150}{300} = \frac{1}{2} = \sin 30°$

$\Rightarrow \theta = 30°$

Now, $AB = \sqrt{300^2 - 150^2} \approx 260$mm

Now taking moment about point A, taking body is in equilibrium

$P \times 300 - 5 \times 260 = 0$

$\Rightarrow 300P = 5 \times 260$

$\Rightarrow P = \frac{5\times 260}{300} = 4.33$kN (Ans)

Calculation the reaction of block

Let R = Reaction of block

As the body is in equilibrium, by resolving all the force in horizontal direction and then equate to zero,

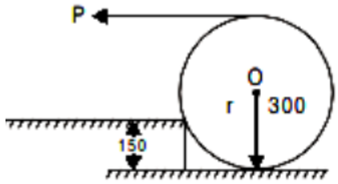
∴ $R\cos 30° - P\sin 30° = 0$

$\Rightarrow \frac{\sqrt{3}R}{2} - \frac{4.33}{2} = 0$

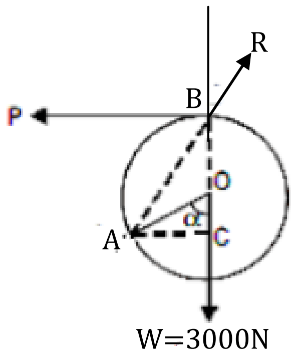
$\Rightarrow \frac{\sqrt{3}R}{2} = \frac{4.33}{2}, \Rightarrow R = \frac{4.33}{\sqrt{3}} = 2.5$kN (Ans)

{Prepared by NatiTute YouTube Channel}

22. A uniform wheel of 600mm diameter, weighing 3kN is to be pulled over a rigid rectangular block of height 150mm by a horizontal force *P* applied to the end of a string wound around the circumference of the wheel shown in the figure. Find the least pull required just to turn the wheel over the corner of the block.



Solution:



From the given condition and free body diagram above we can say, OC=150mm, AO=300mm

$\therefore \cos\alpha = \frac{150}{300} = \frac{1}{2} = \cos 60°$, $\Rightarrow \alpha = 60°$

$\therefore \angle ABC = \frac{60}{2} = 30°$

Now, it is clear that,

Angle between P and W is = 90°

Angle between P and R is = 90° + 30 = 120°

Angle between R and W is = 180° − 30° = 150°

Applying Lami's theorem we get,

$\frac{P}{\sin 150°} = \frac{R}{\sin 90°} = \frac{W}{\sin 120°}$

$\Rightarrow \frac{P}{\sin 150°} = \frac{3000}{\sin 120°}$

$\Rightarrow P = \frac{3000 \times \sin 150°}{\sin 120°} = 1732.05\text{ N}$ (Ans)

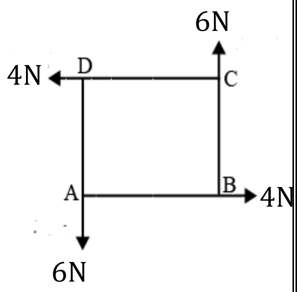
Again,

$\frac{R}{\sin 90°} = \frac{W}{\sin 120°}$

$\Rightarrow R = \frac{3000}{\sin 120°} = 3464.10\text{ N}$ (Ans)

23. Forces each of 4N in magnitude act along AB and CD of square ABCD and forces each of 6N in magnitude act along the sides AD and CB respectively in the same square. Find the moment of the couple which will produce equilibrium. Length of each sides of the square is 80mm.

Solution:



Length of each side of the square

AB=BC=CD=DA=80mm

=0.08m

Moments of couples along

BC || DA = 4×0.08

=0.32 Nm

Moments of couples along

AB || CD = 6×0.08

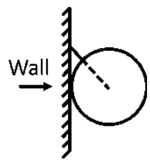
=0.48 Nm

∴ Resultant Moment of couples

=0.32+0.48

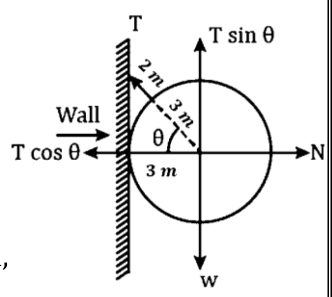
=0.8 Nm (Ans)

24. A uniform sphere of weight W and radius $3m$ is being held by a string of length $2m$. attached to a frictionless wall as shown in the figure. Find the tension in the string.



Solution:

By the free body diagram



$\cos\theta = \frac{3}{5}$

$\Rightarrow \sin\theta = \sqrt{1 - \left(\frac{3}{5}\right)^2}$

$= \frac{4}{5}$

As sphere is at equilibrium,

$\therefore T\sin\theta = W$

$\Rightarrow T \times \frac{4}{5} = W$

$\Rightarrow T = \frac{5W}{4}$ (Ans)

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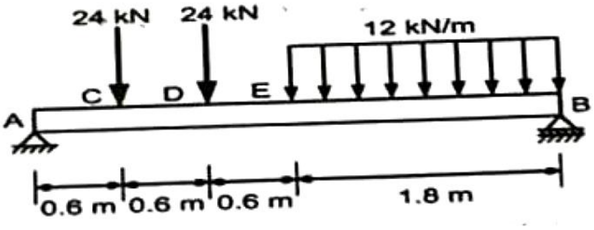
25. Determine the reactions of the support A & B in the adjacent Fig.



Answer:

Let, the support reactions are R_A and R_B.

$\therefore R_A + R_B = 24 + 24 + 12 \times 1.8 = 69.6 \text{ kN}$

$\Rightarrow R_A + R_B = 69.6 \text{ kN}$

Now taking moments of all forces with respect to point B and equating to zero we get,

$$(R_A \times 3.6) - (24 \times 3.0) - (24 \times 2.4) - \left(12 \times 1.8 \times \frac{1.8}{2}\right) + (R_B \times 0) = 0$$

$\Rightarrow (R_A \times 3.6) - 72 - 57.6 - 19.44 = 0$

$\Rightarrow (R_A \times 3.6) = 149.04$

$\Rightarrow R_A = \frac{149.04}{3.6} = 41.4 \text{ kN}$ (Ans)

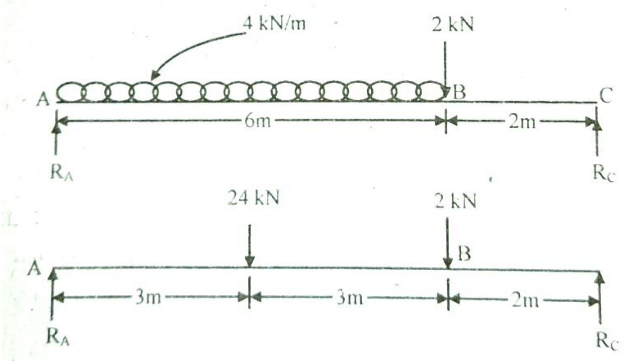
$\because R_A + R_B = 69.6$

$\Rightarrow 41.4 + R_B = 69.6$

$\Rightarrow R_B = 69.6 - 41.4 = 28.2 \text{ kN}$ (Ans)

26. A simply supported beam has a span of 8.0m length. It is subjected to uniformly distributed loading of 4kN/m from left support up to 6.0m length. It is also subjected to a concentrated load of 2kN at a distance of 2.0m from right support. Find the support reaction for this beam.

Solution:



From above diagram we can say,

Total load due to UDL is = 4×6=24kN

We can consider the UDL acting on middle of AB

$\therefore R_A + R_C = 24 + 2 = 26\text{kN}$

Taking moment with respect to point A we get,

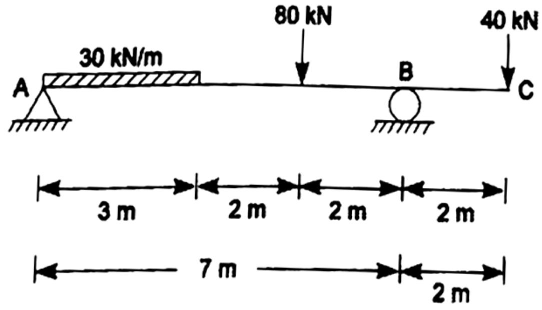
$(R_A \times 0) - (24 \times 3) - (2 \times 6) + (R_C \times 8) = 0$

$\Rightarrow 8R_C - 72 - 12 = 0$

$\Rightarrow 8R_C = 84, \Rightarrow R_C = \frac{84}{8} = 10.5\text{kN}$ (Ans)

$\therefore R_A = 26 - R_C = 26 - 10.5 = 15.5\text{kN}$ (Ans)

27. Find out the reactions at supports by equation of statics for beam of figure below.



Solution:

Let, reactions at A and B are R_A and R_B.

Total UDL = 30×3=90kN,

acts at middle point of UDL.

$R_A + R_B = 90 + 80 + 40 = 210\text{kN}$

Taking moment with respect to point A we get,

$$(R_A \times 0) - \left(90 \times \frac{3}{2}\right) - (80 \times 5) + (R_B \times 7) - (40 \times 9) = 0$$

$\Rightarrow 7R_B - 135 - 400 - 360 = 0$

$\Rightarrow 7R_B = 895$

$\Rightarrow R_B = \frac{895}{7} = 127.86\text{kN}$ (Ans)

$\therefore R_A = 210 - 127.86 = 82.14\text{kN}$ (Ans)

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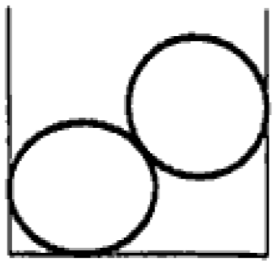
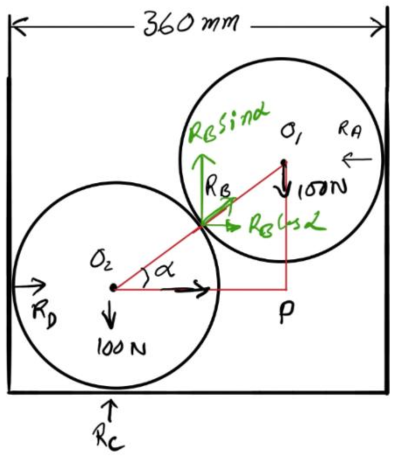
28. Two smooth spheres each of radius 100mm and weighing 100N, rest in a channel having smooth vertical walls, the distance between which is 360 mm. Find the reactions at the points of contacts.



Solution: Reactions are shown in the diagram below-



$O_1O_2 = 100 + 100 = 200\ mm$

$PO_2 = 360 - 100 - 100 = 160\ mm$

$PO_1 = \sqrt{200^2 - 160^2} = 120\ mm$

$\cos\alpha = \frac{PO_2}{O_1O_2} = \frac{160}{200} = 0.8$

$\sin\alpha = \frac{PO_1}{O_1O_2} = \frac{120}{200} = 0.6$

Sphere 1:

$\sum V = 0$ Gives,

$R_B \sin\alpha = 100, \Rightarrow R_B = \frac{100}{0.6} = 166.67$ N (Ans)

$\sum H = 0$ Gives,

$R_A = R_B \cos\alpha = 166.67 \times 0.8 = 133.33$ N (Ans)

Sphere 2:

$\sum V = 0$ Gives,

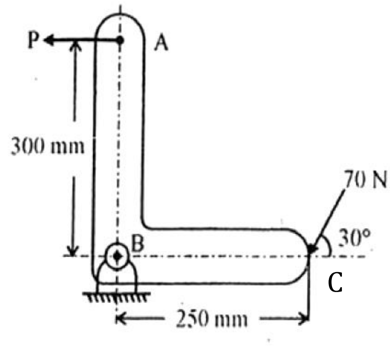
$R_C = 100 + R_B \sin\alpha$

$= 100 + 166.67 \times 0.6 = 200$ N (Ans)

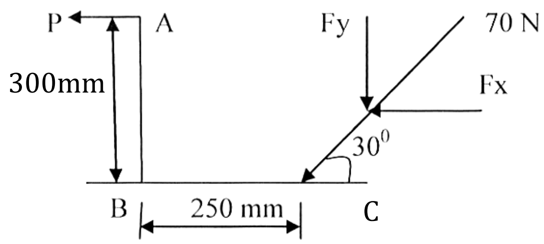
$\sum H = 0$ Gives,

$R_D = R_B \cos\alpha = 166.67 \times 0.8 = 133.33$ N (Ans)

29. A crank ABC with system of forces acting on it is shown in figure. Find force "P' to maintain equilibrium.



Solution:



Given, 70 N force acting at 30° inclination as shown Find : P, if equilibrium is maintained.

Taking moments of all forces with respect to point B we get,

$P \times 300 - F_y \times 250 = 0$

$\Rightarrow 300P = 250F_y$

$\Rightarrow 300P = 250 \times 70 \sin 30°$

$\Rightarrow P = \frac{250\times 70 \sin 30°}{300} = 29.17$N (Ans)

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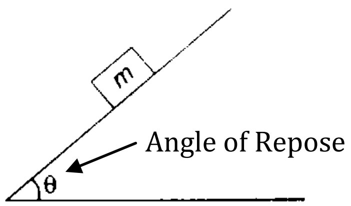
1. **The magnitude of frictional force depends upon- a) area b) shape c) nature of the contacting surfaces d) none of these.**
   Explanation: The magnitude of the frictional force primarily depends on the nature of the contacting surfaces, such as their roughness and the type of material. It is also influenced by the normal force between the surfaces.
   Answer: (c)

2. **The angle which an inclined surface makes with the horizontal when a body placed on it is on the point of moving down is known as –**
   (a) angle of friction
   (b) angle of limiting friction
   (c) angle of repose
   (d) angle of inclination
   Explanation: Angle of repose is the minimum angle that an inclined plane makes with the horizontal when a body placed on it just begins to slide down.



   Answer: (c)

3. **When a body is about to move upwards in an inclined plane by application of force the frictional force will act-**
   (a) upward the inclined plane
   (b) downward the inclined plane
   (c) in the direction of applied force
   (d) none of these
   Explanation: As an object is moving upwards the direction of friction will be in downward direction. Because, the direction of friction force always acts in a direction opposite to which the body tends to move.
   Answer: (b)

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4. **The magnitude of the force of friction between two bodies are laying above the other, depends upon the roughness of the-**
   (a) upper body
   (b) lower body
   (c) both the bodies
   (d) none of these
   Explanation: Friction depends on the nature of the two surfaces. It depends on smoothness or roughness of the two surfaces which are in contact of each other.
   Answer: (c)

5. **Kinetic friction is**
   (a) Maximum value of frictional force when a body is about to move
   (b) Friction between two well lubricated bodies
   (c) Friction force acting when the body is in motion
   (d) Friction force which keeps a body in motion
   Explanation: Kinetic friction is defined as a force that acts between moving surfaces. A body moving on the surface experiences a force in the opposite direction of its movement.
   Answer: (c)

6. **A heavy ladder resting on floor and against a vertical wall may not be in equilibrium, if**
   (a) **the floor is smooth, the wall is rough**
   (b) **the floor is rough, the wall is smooth**
   (c) **the floor and wall both are smooth surfaces**
   (d) **the floor and wall both are rough surfaces**
   Explanation:
   For smooth surface coefficient of friction $\mu = 0$
   Therefore frictional force $F = \mu R = 0 \times R = 0$
   Answer: (c)

7. **The coefficient of friction depends on**
   (a) area of contact
   (b) shape of surfaces
   (c) strength of surfaces
   (d) nature of surface
   Explanation: The coefficient of friction only depends on the nature of the surfaces. It does not depend on any other factors, including the relative speed of the surfaces and the surface area of contact.
   Answer: (a)

8. **On a ladder resting on smooth ground and leaning against vertical wall, the force of friction will be**
   (a) towards the wall at its upper end
   (b) away from the wall at its upper end
   (c) upwards at its upper end
   (d) downwards at its upper end
   Explanation: As the horizontal ground is smooth and the vertical wall is rough. The ladder moves horizontally, the upper end of the ladder starts moving down so the frictional force will act upward at its upper end.
   Answer: (c)

9. **Define the term friction. How does it come into play?**
   Answer:
   Friction: Friction is defined as the resistance offered by the surfaces that are in contact when they move past each other.
   Friction is caused due to the irregularities on the two surfaces in contact. So, when one object moves over the other, these irregularities on the surface get entangled, giving rise to friction. The more the roughness, the more irregularities and more significant will be the friction.

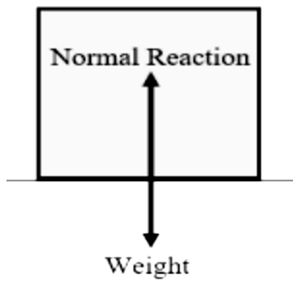
10. **What is static friction?**
    Answer:
    Static Friction: Static friction is defined as the frictional force that acts between the surfaces when they are at rest with respect to each other.

11. **What is sliding friction?**
    Answer:
    Sliding Friction: Sliding friction is defined as the resistance that is created between any two objects when they are sliding against each other.

12. **Define rolling friction.**
    Answer:
    Rolling Friction: Rolling friction is defined as the force which resists the motion of a ball or wheel and is the weakest type of friction.

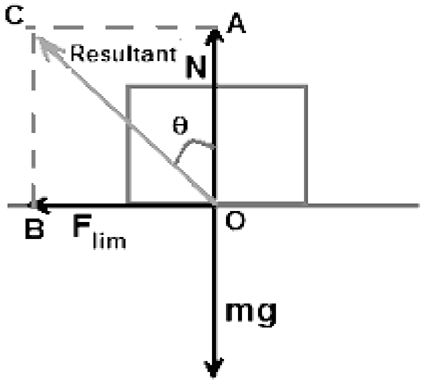
13. **What is fluid friction?**
    Answer:
    Fluid Friction: Fluid friction is defined as the friction that exists between the layers of the fluid when they are moving relative to each other.

14. **Define normal reaction.**
    Answer:
    Normal Reaction: The normal reaction is defined as the force that any surface exerts on any other object. If that object is at rest, then the net force acting on the object is equal to zero. It is a fact that the downward force i.e. weight must be equal to the upward force i.e. the normal reaction.



15. **Explain the term co-efficient of friction.**
    Answer:
    Co-efficient of Friction: A coefficient of friction is a value that shows the relationship between the force of friction (F) between two objects and the normal reaction (R) between the objects that are involved. Mathematically,
    $$F = \mu R$$
    Where R is the normal force. F is friction force and μ is friction coefficient.

16. **Explain the term angle of friction.**
    Answer:
    Angle of Friction: Angle made by the resultant of normal reaction and limiting friction (F_{lim}) with the normal reaction (N) is called the angle of friction (θ). The coefficient of static friction is equal to the tangent of the angle of friction. i.e $\mu = \tan\theta$



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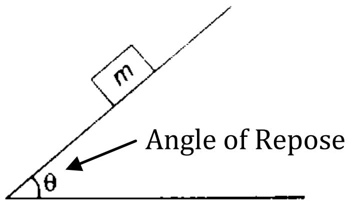
17. Define Angle of Repose.

Answer:

Angle of Repose: Angle of repose is the minimum angle that an inclined plane makes with the horizontal when a body placed on it just begins to slide down.



18. Prove that angle of repose equal to the angle of friction.

Answer: The expression for angle of friction is given as $\tan\phi = \mu$

Here ϕ is the angle of friction and μ is the coefficient of limiting friction.

The expression for angle of repose is given as

$$\tan\theta = \mu$$

Here θ is the angle of repose and μ is the coefficient of limiting friction.

Since both tangent of the angle of friction and the tangent of the angle of repose are numerically equal to each other, we can say $\tan\phi = \tan\theta$

$$\therefore \phi = \theta$$

Thus it is proved that angle of friction and angle of repose are equal.

19. State laws of static friction.

Answer:

Laws of Friction: There are five laws of friction and they are:

i) The friction of the moving object is proportional and perpendicular to the normal force.

ii) The friction experienced by the object is dependent on the nature of the surface it is in contact with.

iii) Friction is independent of the area of contact as long as there is an area of contact.

iv) Kinetic friction is independent of velocity.

v) The coefficient of static friction is greater than the coefficient of kinetic friction.

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20. What is limiting friction?

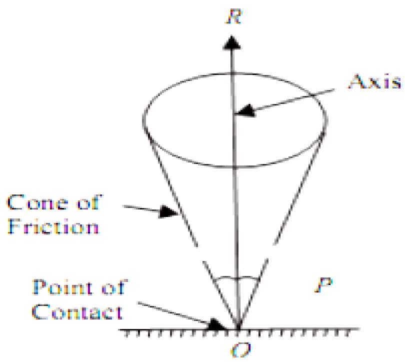
Answer:

Limiting Friction: The maximum static friction that a body can exert on the other body in contact with it is called limiting friction. The limiting frictional force is independent of the area of contact and is proportional to the reasonable reaction between the contacting surfaces.

21. What is cone of friction?

Answer:

Cone of Friction: It can be defined as the right circular cone having vertex at the point of contact of two bodies (or the surfaces), axis in direction of normal reaction (R) and semi vertical angle equal to the angle of friction (θ).



22. What are the advantages of friction?

Answer:

Advantages of Friction are-

i) Friction enables us to walk freely.

ii) It helps to support ladder against wall.

iii) It becomes possible to transfer one form of energy to another.

iv) Objects can be piled up without slipping.

v) Breaks of vehicles work due to friction.

vi) Friction causes asteroids or stones to disintegrate throughout the atmosphere.

23. What are the disadvantages of friction?

Answer:

Disadvantages of Friction are-

i) A lot of heat is generated by friction in various mechanical parts, which wastes energy as heat.

ii) Restricts motion, requiring more energy to overcome it.

iii) Making noise while operating a machine wastes energy and is annoying.

iv) Friction between tree branches is what ignites forest fires.

v) It causes wear and tear of machines.

24. The force required to pull a body of weight 60N on a rough horizontal plane is 15N. Determine the coefficient of friction if the force is applied at an angle of 15° with the horizontal.

Solution:

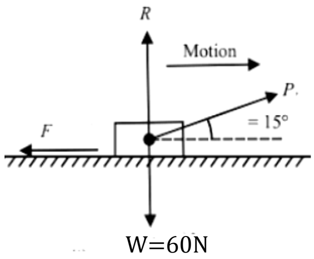
From the given condition

$P = 15\text{N}$

Horizontal component of P is $P\cos 15°$

$\therefore F = 15\cos 15°$

$= 14.49\text{N}$



Vertical component

$R = W - P\sin\theta$

$\Rightarrow R = 60 - 15\sin 15° = 56.12\text{N}$

Again, $F - \mu R = 0$ [μ = coefficient of friction]

$\Rightarrow \mu = \frac{F}{R} = \frac{14.49}{56.12} = 0.26$ (Ans)

25. A body of weight 500N is lying on a rough plane inclined at an angle of 25° with the horizontal. It is supported by an effort (P) parallel to the plane. Determine the minimum value of P, for which the equilibrium can exist, if the angle of friction is 20°.

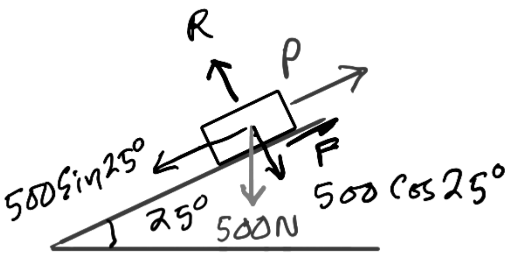
Solution: Given data-

Weight of the body: W=500N

Inclination angle: θ=25°

Angle of friction: ϕ=20°

The force system can be drawn as follows-



Here, $R = 500\cos 25° = 453.15$ N

Coefficient of friction $\mu = \tan\phi = \tan 20° = 0.36$

$\therefore$ Frictional force

$$F = \mu R = 0.36 \times 453.15 = 163.13 \text{ N}$$

For equilibrium, the sum of forces parallel to the incline must be zero:

$P + F - W\sin\theta = 0$

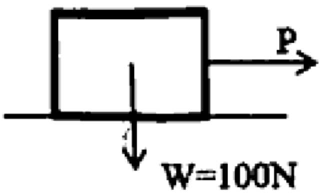
$\Rightarrow P = W\sin\theta - F$

$\Rightarrow P = 500\sin 25° - 163.13$

$\Rightarrow P = 211.31 - 163.13 = 48.18$ N

$\therefore$ The minimum effort required for equilibrium is 48.18 N (Ans)

26. Find the friction force developed in the given figure for P = 20N, 50N and 60N. Coefficient of static friction μ_s= 0.5 and Coefficient of kinetic (or dynamic) friction μ_d = 0.4.



Solution: Given, $W = 100N$

$\therefore$ Normal reaction $R = W = 100N$

Coefficient of static friction $\mu_s = 0.5$

Coefficient of dynamic friction $\mu_d = 0.4$

$\therefore$ Maximum static friction force

$F_s = \mu_s R = 0.5 \times 100 = 50$ N

Maximum dynamic friction force

$F_d = \mu_d R = 0.4 \times 100 = 40$ N

Case 1: For P =20N

P is less than the maximum static friction (F = 50N).

The body remains at rest, and the friction force adjusts to match P.

Friction force (Static) = 20N (Ans)

Case 2: For P = 50N

P is equal to the maximum static friction.

The body is on the verge of motion.

Friction force (Static) = 50N (Ans)

Case 3: For P = 60N

P exceeds the maximum static friction.

The body starts moving, and kinetic friction takes over.

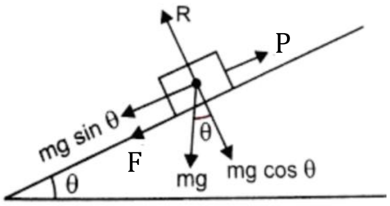
As kinetic friction remains constant once motion starts.

Friction force (dynamic) = 40N (Ans)

{Prepared by NatiTute YouTube Channel}

27. A weight of 25N is pulled up a rough inclined plane whose inclination to the horizontal is 30° by a force of 18N acting parallel to the plane. Calculate the coefficient of friction.

Solution:



From the given condition,

$mg = 25\text{N}, \theta = 30°, P = 18\text{N}$

$\therefore$ Normal reaction $R = mg\cos\theta = 25\cos 30°$

$= 21.65\text{N}$

Net pulling force $P = F + mg\sin\theta$

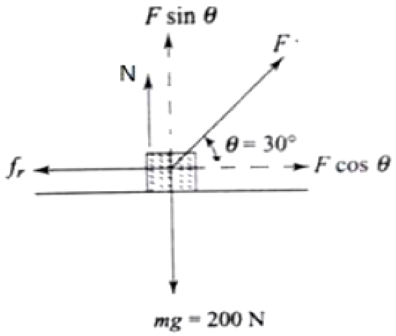
$\Rightarrow 18 = F + 25\sin 30°$

$\Rightarrow F = 18 - 25\sin 30° = 5.5\text{N}$

$\therefore$ Coefficient of friction $\mu = \frac{F}{R} = \frac{5.5}{21.65} = 0.25$(Ans)

28. A metal block of weight 200N is resting on a horizontal surface. The coefficient of friction between the surfaces being 0.3. A pull is exerted on the block by means of a string inclined at 30° to the horizontal which is sufficient to cause motion of the block. Calculate the magnitude of force exerted by the string.

Solution:



Since the block is resting, no net force acts on it. Therefore, the horizontal component Fcosθ of force F must balance the frictional force force, i.e., $f_r = F\cos\theta$.

Now, $N + F\sin\theta = mg$

$\Rightarrow N = mg - F\sin\theta$

Also, $f_r = \mu N$

$\Rightarrow F\cos\theta = \mu(mg - F\sin\theta)$

$\Rightarrow F\cos 30° = 0.3(200 - F\sin 30°)$

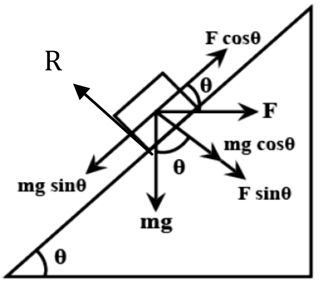
$\Rightarrow 0.87F = 60 - 0.5F$

$\Rightarrow 1.37F = 60$

$\Rightarrow F = \frac{60}{1.37} = 47.80\text{N}$ (Ans)

29. A load W acts on an inclined plane of inclination θ to the horizontal. If the coefficient of friction μ, show that the horizontal necessary to move the load up the plane is $\frac{\mu + \tan\theta}{1 - \mu\tan\theta} \times W$.

Solution:



The body is in equilibrium under the action of forces shown in above figure.

Resolving the forces along the plane

$F\cos\theta = W\sin\theta + \mu R$ ...(i) $[W = mg]$

Resolving the forces normal to the plane

$R = F\sin\theta + W\cos\theta$

Substituting the value of R in equation (i)

$F\cos\theta = W\sin\theta + \mu(F\sin\theta + W\cos\theta)$

$\Rightarrow F\cos\theta = W\sin\theta + \mu F\sin\theta + \mu W\cos\theta$

$\Rightarrow F\cos\theta - \mu F\sin\theta = W\sin\theta + \mu W\cos\theta$

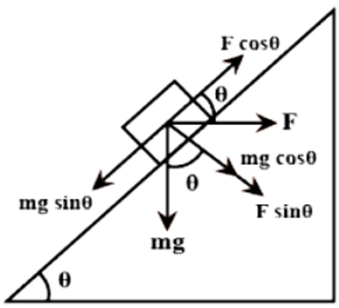
$\Rightarrow F(\cos\theta - \mu\sin\theta) = W(\sin\theta + \mu\cos\theta)$

$\Rightarrow F = \frac{\sin\theta + \mu\cos\theta}{\cos\theta - \mu\sin\theta} \times W$

$\Rightarrow F = \frac{\mu + \tan\theta}{1 - \mu\tan\theta} \times W$ (Proved)

30. Determine the horizontal force F to be applied to a block of weight 450N to hold it in position on a smooth inclined plane which makes an angle 30° with horizontal.

Solution:



According to the given codition,

weight $mg = 450\text{N}, \theta = 30°$

Resolving the forces along the plane

$F\cos\theta = mg\sin\theta$ [As the plane is smooth]

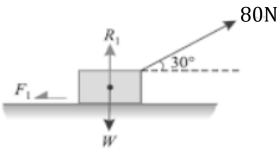
$\Rightarrow F\cos 30° = 450\sin 30°$

$\Rightarrow F = \frac{450\sin 30°}{\cos 30°} = 259.81\text{N}$

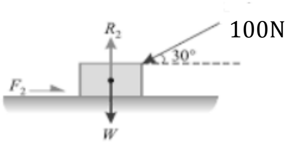
$\therefore$ The horizontal force applied $= 259.81\text{N}$ (Ans)

31. A body resting on a horizontal plane required a pull of 80N inclined to 30° to horizontal just to move it. It was also found that a push of 100N inclined at 30° to horizontal just to move the body. Find the weight of the body and coefficient of friction.

Solution:



Pulled



Pushed

From the above diagram,

Resolving the forces when the block is pulled,

$F_1 = 80\cos 30° = 69.28\text{N}$

$R_1 = W - 80\sin 30° = (W - 40)\text{N}$

From formula $F = \mu R$ we get,

$F_1 = \mu R_1, \Rightarrow 69.28 = \mu(W - 40)$ …….. (i)

Resolving the forces when the block is pushed,

$F_2 = 100\cos 30° = 86.60\text{N}$

$R_2 = W + 100\sin 30° = (W + 50)\text{N}$

From formula $F = \mu R$ we get,

$F_2 = \mu R_2, \Rightarrow 86.60 = \mu(W + 50)$ …….. (ii)

Dividing (i) by (ii) we get,

$\frac{69.28}{86.60} = \frac{\mu(W-40)}{\mu(W+50)}$

$\Rightarrow 86.60W - 3464 = 69.28W + 3464$

$\Rightarrow 17.32W = 6928$

$\Rightarrow W = \frac{6928}{17.32} = 400\text{N}$ (Ans)

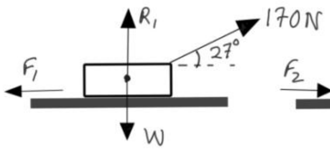
Now, substituting the value of W in equation (i)

$69.28 = \mu(400 - 40)$

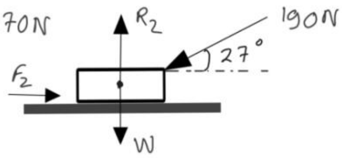
$\Rightarrow \mu = \frac{69.28}{360} = 0.192$ (Ans)

32. A body is resting on rough horizontal plane. It requires a pull of 170N inclined plane at 27°, to the plane just to move it, while a push of 190N inclined at 27° to the plane just to move it. Determine weight of the body and the coefficient of friction.

Solution:



Pulled



Pushed

From the above diagram,

Resolving the forces when the block is pulled,

$F_1 = 170\cos 27° = 151.47\text{N}$

$R_1 = W - 170\sin 27° = (W - 77.18)\text{N}$

From formula $F = \mu R$ we get,

$F_1 = \mu R_1, \Rightarrow 151.47 = \mu(W - 77.18)$ …….. (i)

Resolving the forces when the block is pushed,

$F_2 = 190\cos 27° = 169.29\text{N}$

$R_2 = W + 190\sin 27° = (W + 86.26)\text{N}$

From formula $F = \mu R$ we get,

$F_2 = \mu R_2, \Rightarrow 169.29 = \mu(W + 86.26)$ …….. (ii)

Dividing (i) by (ii) we get,

$\frac{151.47}{169.29} = \frac{\mu(W-77.18)}{\mu(W+86.26)}$

$\Rightarrow \frac{W-77.18+W+86.26}{W-77.18-W-86.26} = \frac{151.47+169.29}{151.47-169.29}$

$\Rightarrow \frac{2W+9.08}{-163.44} = \frac{320.76}{-17.82}$

$\Rightarrow 2W + 9.08 = \frac{320.76\times163.44}{17.82} = 2941.92$

$\Rightarrow 2W = 2941.92 - 9.08 = 2932.84$

$\Rightarrow W = 1466.42\text{ N}$ (Ans)

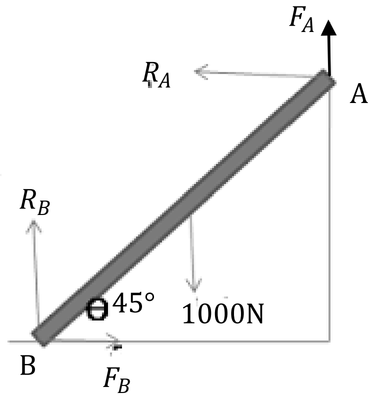
Now, substituting the value of W in equation (i)

$151.47 = \mu(1466.42 - 77.18)$

$\Rightarrow \mu = \frac{151.47}{1466.42-77.18} = 0.109$ (Ans)

33. A uniform ladder of weight 1000N, inclined to the horizontal at 45°, rests with its upper extremity against a rough vertical wall and its lower extremity on the ground. Find the least horizontal force which will move the lower end towards the wall. Given 0.25 and 0.20 are the coefficients of friction at the lower and upper ends respectively.

Solution:



Let, length of the ladder $= L$

Taking moment of all forces with respect to A,

$$R_B \times L\sin 45° - F_B \times L\cos 45° - 1000 \times \frac{L}{2}\cos 45° = 0$$

$\Rightarrow \frac{R_B}{\sqrt{2}} - \frac{0.25R_B}{\sqrt{2}} - \frac{500}{\sqrt{2}} = 0$

$\Rightarrow 0.75R_B = 500$

$\Rightarrow R_B = \frac{500}{0.75} = 666.67\text{N}$

Considering vertical forces,

$F_A + R_B = 1000$

$\Rightarrow F_A + 666.67 = 1000$

$\Rightarrow F_A = 333.33\text{ N}$

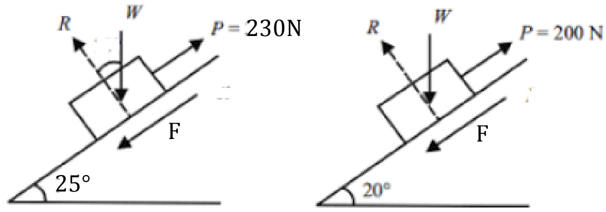
$\therefore R_A = \frac{F_A}{\mu_A} = \frac{333.33}{0.20} = 1666.65\text{N}$

$\therefore$ Least horizontal force required

$R_A = F_B = 1666.65\text{N}$ (Ans)

34. A force of 200N is required just to move a certain body up an inclined plane of angle 20°, the force is acting parallel to the plane. If the angle of inclination of the plane is made 25°, the force required parallel to plane is 230N. Find the coefficient of friction and weight of the body.

Solution:



Case-1:

Consider when the slope of the plane be 25°

Sum of forces parallel to plane

$230 - \mu R - W\sin 25° = 0$ ..... (i)

Sum of forces perpendicular to plane

$R = W\cos 25°$ .... (ii)

Putting the value of R in (i) We get

$230 - \mu W\cos 25° - W\sin 25° = 0$

Dividing both side by $\cos 25°$

$\Rightarrow 253.78 - \mu W - 0.47W = 0$ ..... (iii)

Case-2:

Similarly for case 2 we get,

$200 - \mu W\cos 20° - W\sin 20° = 0$

Dividing both side by $\cos 20°$

$\Rightarrow 212.83 - \mu W - 0.36W = 0$ ..... (iv)

Subtracting (iv) from (iii) we get,

$40.95 - 0.11W = 0$

$\Rightarrow W = \frac{40.95}{0.11} = 372.27\text{N}$ (Ans)

Now from equation (iii) we can write,

$253.78 - 372.27\mu - 0.47 \times 372.27 = 0$

$\Rightarrow 372.27\mu = 78.81$

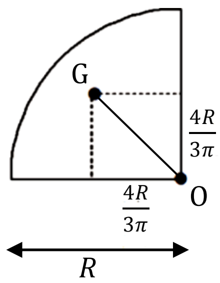
$\Rightarrow \mu = \frac{78.81}{372.27} = 0.212$ (Ans)

{Prepared by NatiTute YouTube Channel}

1. **The point, through which the whole weight of the body acts, irrespective of its position, is known as (a) Moment of inertia (b) Center of gravity (c) Center of percussion (d) Center of mass.**
Explanation: The point where the total weight of the body focuses upon is known as the centre of gravity. It is the point where the gravitational force (weight) acts on the body, it is denoted by g or CG.
Answer: (b)

2. **The CG of solid circular cone divides the height in the ratio-**
(a) 1:2 (b) 2:3 (c) 1:3 (d) 1:4
Explanation: Centre of gravity of right circular cone lies at a distance of $\frac{h}{4}$ from the base and at a distance of $\frac{3h}{4}$ from the apex. So ratio of divisions is $\frac{h}{4} : \frac{3h}{4} = 1 : 3$
Answer: (c)

3. **The centre of gravity of a quadrant of a circle lies along its central radius at a distance of-**
(a) 0.2R
(b) 0.4R
(c) 0.3R
(d) 0.6R
Explanation:



From the figure we can say, The centre of gravity along its central radius

$$OG = \sqrt{\left(\frac{4R}{3\pi}\right)^2 + \left(\frac{4R}{3\pi}\right)^2}$$
$$= \frac{4R}{3\pi} \times \sqrt{2}$$
$$= 0.60$$

Answer: (d)

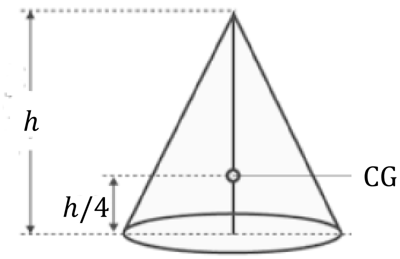
4. **In a half filled circular cylindrical vessel the centre of gravity will be at ___ compared to the same vessel when it was empty.**
(a) higher
(b) same height
(c) lower
(d) none of these
Explanation: Since liquid is denser than air and is located below the original CG, the new CG will be lower than that of the empty vessel.
Answer: (c)

5. **What is the centre of gravity of the cone base 10cm and height 50 cm?**
(a) 10.5 cm
(b) 12.5 cm
(c) 11.25 cm
(d) 12.75 cm
Explanation:
CG of right circular cone $= \frac{h}{4} = \frac{50}{4} = 12.5$ cm
Answer: (b)

6. **The center of gravity of a triangle lies at the point of**
(a) concurrence of the medians
(b) intersection of its altitudes
(c) intersection of bisector of angles
(d) all of the above
Explanation: Hence for triangular lamina centre lies at the centroid which is the intersection of the three lines drawn from the vertex to the midpoint of the opposite side. Hence centre of gravity lies at the intersection of the three medians.
Answer: (a)

7. **To find out the centroid of the given triangular area (height=90, base=60), $\bar{y}$ =?**
(a) 30 (b)20 (c) 45 (d) 15
Explanation: Height of centroid of any triangle $= \frac{1}{3} \times$ Height of triangle$= \frac{1}{3} \times 90 = 30$
Answer: (a)

8. **The CG of a right circular cone of height *h* lies at a distance of ____ from the base.**
(a) $\frac{h}{2}$ (b) $\frac{h}{3}$ (c) $\frac{h}{4}$ (d) $\frac{h}{6}$
Explanation: Centre of gravity of right circular cone lies at a distance of $\frac{h}{4}$ from the base and at a distance of $\frac{3h}{4}$ from the apex.



Answer: (c)

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9. Define centre of gravity.

Answer:

Centre of Gravity: The point where the total weight of the body focuses upon is known as the centre of gravity. It is the point where the gravitational force (weight) acts on the body, it is denoted by g or CG.

10. Define centroid.

Answer:

Centroid: A centroid is the geometrical centre of a body. It refers to the centre of gravity of uniform density objects.

11. Differentiate between centroid and C.G.

Answer: The differences between centroid and centre of gravity are-

i) The centroid is the geometric center of a shape, where the entire area appears to be concentrated. In contrast, the center of gravity (C.G.) is the point where the entire weight of a body is considered to act.

ii) The centroid depends only on the shape and dimensions, while the C.G. depends on both shape and mass distribution.

iii) The centroid exists for all geometric figures, even if they have no mass, whereas the C.G. exists only for objects with mass.

iv) Gravity does not affect the centroid, but it directly influences the C.G.

12. What is the difference between centre of gravity and centre of mass?

Answer:

Difference between centre of gravity and centre of mass: The center of mass (CM) is the point where the entire mass of an object is considered to be concentrated for motion analysis, independent of gravity. The center of gravity (CG), on the other hand, is the point where the total weight of the object acts and depends on the gravitational field. In a uniform gravitational field, the CM and CG coincide, but in a non-uniform field (e.g., large objects like Earth), they may differ. The CM exists even in space without gravity, while the CG is only meaningful in the presence of gravity. For most everyday objects on Earth, the two points are practically the same.

13. What is the centre of mass?

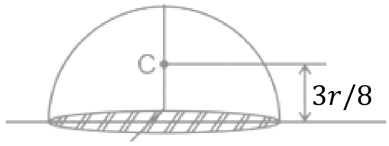
Answer:

Centre of Mass: Centre of mass of a body or system of a particle is defined as, a point at which the whole of the mass of the body or all the masses of a system of particle appeared to be concentrated.

14. A solid hemisphere having radius r then what height of its CG from base?

Answer:

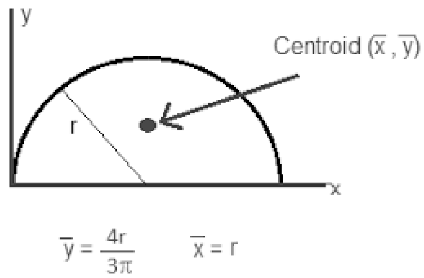
The centre of gravity of solid hemisphere lies on the central radius at a distance of $(\frac{3r}{8})$ from the plane base.



15. What is the height of CG of a semi circular lamina of radius r from its diameter?

Answer:

The centre of gravity of a semi circle lies at a distance of $\frac{4r}{3\pi}$ measured along the vertical radius.



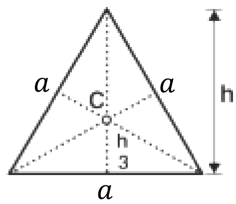
16. Find the CG of an equilateral triangle with each side a.

Answer:

Centre of gravity of any triangle of height h is lies at $\frac{h}{3}$ from base.

For equilateral triangle of side a

$h = \frac{\sqrt{3}}{2}a$



∴ Height of CG of equilateral triangle

$= \frac{h}{3} = \frac{1}{3} \times \frac{\sqrt{3}}{2} a = \frac{a}{2\sqrt{3}}$

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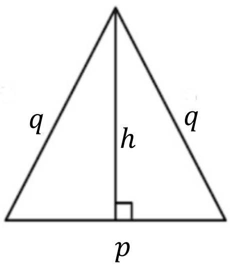
17. Find the centre of gravity of isosceles triangle with base *p* and sides *q* from its base.

Answer:

Let, height of the triangle h

Now from the figure we can say $h^2 + \left(\frac{p}{2}\right)^2 = q^2$



$\Rightarrow h^2 = q^2 - \frac{p^2}{4}$

$\Rightarrow h = \sqrt{q^2 - \frac{p^2}{4}}$

∴ Height of CG of isosceles triangle from its base is

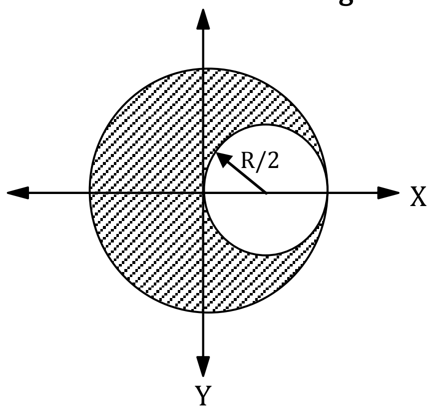
$\frac{h}{3} = \frac{1}{3}\sqrt{q^2 - \frac{p^2}{4}} = \frac{1}{6}\sqrt{4q^2 - p^2}$

18. What do you mean by lamina?

Answer:

Lamina: A lamina is a 2-dimensional object. In other words, it is a flat object whose thickness is we can ignore. If a body has a line of symmetry, the centre of mass will lie on this line.

19. Determine the CG of the section given below.



Solution:

According to the given condition,

a_1 = Area of large circle $= \pi R^2$

a_2 = Area of small circle $= \pi\left(\frac{R}{2}\right)^2 = \frac{\pi R^2}{4}$

x_1 = CG of large circle from origin= 0

x_2 = CG of small circle from origin= $\frac{R}{2}$

∴ CG of whole section from origin

$$\bar{x} = \frac{a_1x_1 - a_2x_2}{a_1 - a_2} = \frac{\pi R^2 \times 0 - \frac{\pi R^2}{4}\times\frac{R}{2}}{\pi R^2 - \frac{\pi R^2}{4}}$$

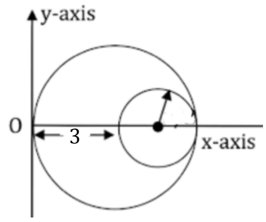
$$= -\frac{\frac{R}{8}}{1-\frac{1}{4}} = -\frac{R}{6}$$

∴ Coordinate of GC $\left(-\frac{R}{6}, 0\right)$

20. From a circular plate of diameter 6cm is cut out a circle whose diameter is a radius of the plate. Find the CG of the remainder from the centre of circular plate.

Solution:

From the given data



$a_1 = \pi \times 3^2 = 9\pi\ cm^2$

$a_2 = \pi \times (1.5)^2 = 2.25\pi\ cm^2$

$x_1 = 3\ cm$

$x_2 = 3 + \frac{3}{2} = 4.5\ cm$

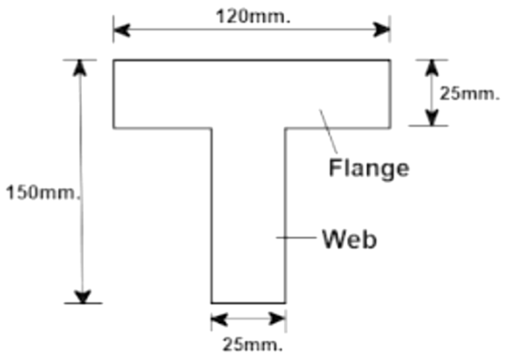
$\therefore \bar{x} = \frac{a_1x_1 - a_2x_2}{a_1 - a_2}$

$= \frac{9\pi\times3 - 2.25\pi\times4.5}{9\pi - 2.25\pi}$

$= \frac{27 - 10.125}{6.75} = 2.5\ cm$

∴ The CG of the remainder from the centre of the circular plate is (3−2.5)=0.5 cm (Ans)

21. Find the centre of gravity of T-section given below.



Solution:

The T-section is split up into two rectangle (1) flange, (2) web. The given T-section is symmetrical about Y-axis. Hence CG of section will lie on this axis. a_1 and a_2 are the area of rectangles and y_1 and y_2 are their respective distance of CG from bottom line.

Now,

a_1 = Area of flange $= 120 \times 25 = 3000\ mm^2$

a_2 = Area of web $= 25 \times 125 = 3125\ mm^2$

$y_1 = 150 - \frac{25}{2} = 137.5\ mm$

$y_2 = \frac{125}{2} = 62.5\ mm$

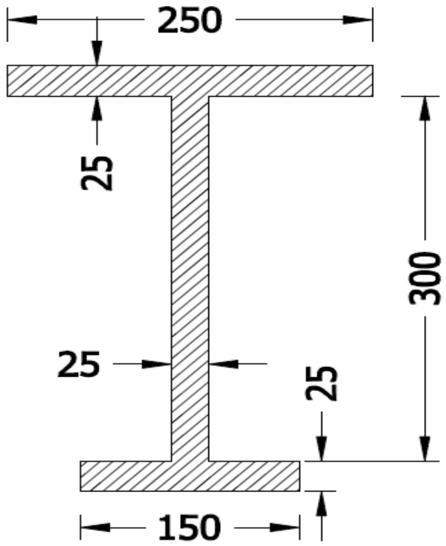
∴ Distance of CG of the T-section from bottom line

$$\bar{y} = \frac{a_1y_1 + a_2y_2}{a_1 + a_2} = \frac{3000\times137.5 + 3125\times62.5}{3000 + 3125}$$

$= 99.23\ mm$ (Ans)

{Prepared by NatiTute YouTube Channel}

22. Find the centroid of the cross section shown in figure. All dimensions are in mm.



Solution:

The I-section is split up into three rectangle (1) top, (2) middle, (3) bottom. The given I-section is symmetrical about Y-axis. Hence CG of section will lie on this axis. a_1, a_2 and a_3 are the area of rectangles and y_1, y_2 and y_3 are their respective distance of CG from bottom line.

Now,

$a_1 = 250 \times 25 = 6250\, mm^2$

$a_2 = 25 \times 300 = 7500\, mm^2$

$a_3 = 150 \times 25 = 3750\, mm^2$

And,

$y_1 = 25 + 300 + \frac{25}{2} = 337.5\, mm$

$y_2 = 25 + \frac{300}{2} = 175\, mm$

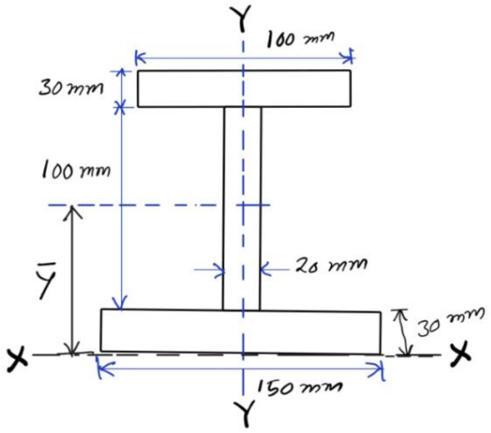
$y_3 = \frac{25}{2} = 12.5\, mm$

∴ Distance of CG of the I-section from bottom line

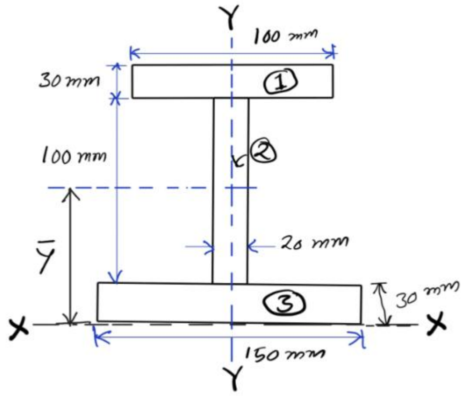
$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3} = \frac{6250 \times 337.5 + 7500 \times 175 + 3750 \times 12.5}{6250 + 7500 + 3750}$$

$= 198.21\, mm$ (Ans)

23. **Locate the centroid of the I-section shown in figure.**



Solution: The I-section is split up into three rectangle (1) top, (2) middle, (3) bottom.



The given I-section is symmetrical about Y-axis. Hence CG of section will lie on this axis. a_1, a_2 and a_3 are the area of rectangles and y_1, y_2 and y_3 are their respective distance of CG from bottom line.

Now,

$a_1 = 100 \times 30 = 3000\, mm^2$

$a_2 = 20 \times 100 = 2000\, mm^2$

$a_3 = 150 \times 30 = 4500\, mm^2$

And,

$y_1 = 30 + 100 + \frac{30}{2} = 145\, mm$

$y_2 = 30 + \frac{100}{2} = 80\, mm$

$y_3 = \frac{30}{2} = 15\, mm$

∴ Distance of CG of the I-section from bottom line

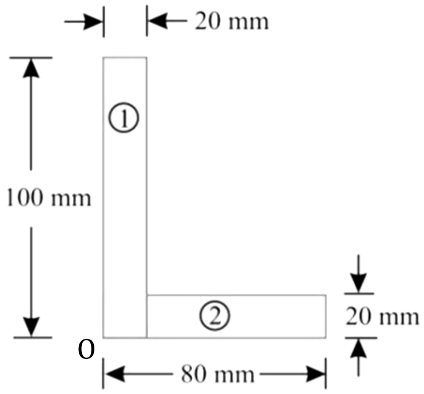
$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3}$$

$$= \frac{3000 \times 145 + 2000 \times 80 + 4500 \times 15}{3000 + 2000 + 4500}$$

$= 69.74\, mm$ (Ans)

24. **Find the CG of an unequal angle section 100mm ×80mm×20mm**

Solution:



The section is split up into two rectangle (1) and (2). The given section is not symmetrical about any axis. a_1 and a_2 are the area of rectangles and y_1 and y_2 are their respective distance of CG from bottom line and x_1 and x_2 are their respective distance of CG from left side.

Now,

$a_1 = 20 \times 100 = 2000\, mm^2$

$a_2 = (80 - 20) \times 20 = 1200\, mm^2$

$y_1 = \frac{100}{2} = 50\, mm\,,\, y_2 = \frac{20}{2} = 10\, mm$

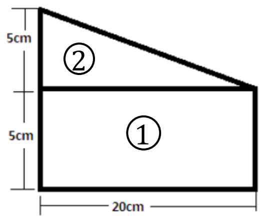
$x_1 = \frac{20}{2} = 10\, mm\,,\, x_2 = 20 + \frac{80-20}{2} = 50\, mm$

$\therefore\ \bar{x} = \frac{a_1x_1+a_2x_2}{a_1+a_2} = \frac{2000\times10+1200\times50}{2000+1200} = 25\, mm$

$\bar{y} = \frac{a_1y_1+a_2y_2}{a_1+a_2} = \frac{2000\times50+1200\times10}{2000+1200} = 35\, mm$

$\therefore$ The coordinate of CG of the angle section, considering (O) as origin, be (25mm, 35mm)

25. **Find the height of CG of the figure shown.**



Solution:

$a_1 = 20 \times 5 = 100\, cm^2$

$a_2 = \frac{1}{2} \times 20 \times 5 = 50\, cm^2$

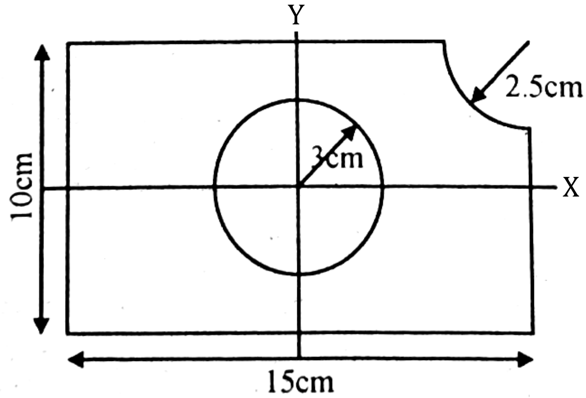
$y_1 = \frac{5}{2} = 2.5\, cm, y_2 = 5 + \frac{5}{3} = 6.66\, cm$

$\therefore$ Height of CG $\bar{y} = \frac{a_1y_1+a_2y_2}{a_1+a_2}$

$= \frac{100\times2.5+50\times6.66}{100+50}$

$= 3.88\, cm$ (Ans)

26. **Find the centroid for the following figure having dimensions as shown.**



Solutions:

According to given figure a circle and a quadrant of circle is cut out from a rectangle.

Now,

$a_1 =$ Area of rectangle $= 15 \times 10 = 150\, cm^2$

Coordinate of centroid $x_1 = 0, y_1 = 0$

$a_2 =$ Area of circle $= \pi \times 3^2 = 28.27\, cm^2$

Coordinate of centroid $x_2 = 0, y_2 = 0$

$a_3 =$ Area of quarter circle $= \frac{\pi}{4} \times (2.5)^2$

$= 4.91\, cm^2$

Coordinate of centroid

$x_3 = \frac{15}{2} - \frac{4\times2.5}{3\pi} = 7.5 - 1.06 = 6.44\, cm$

$y_3 = \frac{10}{2} - \frac{4\times2.5}{3\pi} = 5 - 1.06 = 3.94\, cm$

$\therefore\ \bar{x} = \frac{a_1x_1-a_2x_2-a_3x_3}{a_1-a_2-a_3} = \frac{150\times0-28.27\times0-4.91\times6.44}{150-28.27-4.91}$

$= -0.270\, cm$

$\bar{y} = \frac{a_1y_1-a_2y_2-a_3y_3}{a_1-a_2-a_3} = \frac{150\times0-28.27\times0-4.91\times3.94}{150-28.27-4.91}$

$= -0.165\, cm$

$\therefore$The coordinate of CG of whole area is

$(-0.27cm, -0.165cm)$ (Ans)

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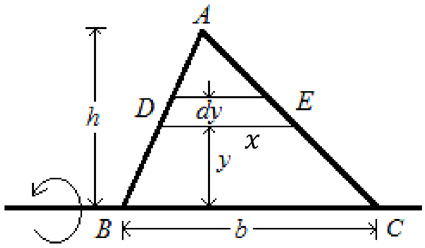
27. Find out the centroid of a uniform triangular lamina by the method of integration.

Solution:



The X-axis is taken to coincide with the base.

A differential strip of area $dA = xdy$ is chosen.

By similar triangle, $\frac{\text{base}}{\text{height}} = \frac{x}{h-y} = \frac{b}{h}$

$$\Rightarrow x = \frac{b}{h}(h-y)$$

Area of the triangle $= \frac{1}{2}bh$

Moment of area $= \int_0^h yx\,dy$

$= \int_0^h y \times \frac{b}{h}(h-y)\,dy$

$= \frac{b}{h}\int_0^h (hy - y^2)\,dy$

$= \frac{b}{h}\left[\frac{hy^2}{2} - \frac{y^3}{3}\right]_0^h$

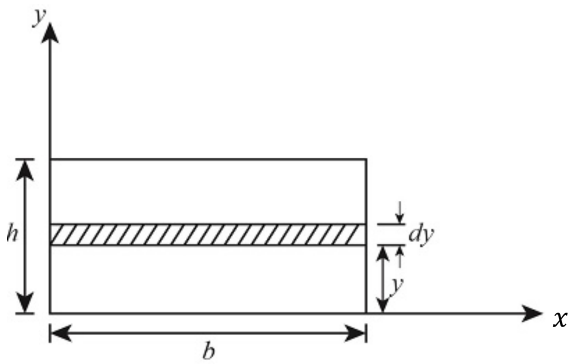
$= \frac{b}{h}\left[\frac{h.h^2}{2} - \frac{h^3}{3}\right] = \frac{bh^3}{h}\left[\frac{1}{2} - \frac{1}{3}\right] = \frac{bh^2}{6}$

$\therefore\ \bar{y} = \frac{\text{Moment of area}}{\text{Total area}} = \frac{\frac{bh^2}{6}}{\frac{1}{2}bh} = \frac{h}{3}$

Thus the centroid of triangle is at a distance $\frac{h}{3}$ from the base of the triangle where h is the height of the trangle.

28. Find out the centroid of a uniform rectangular lamina by the method of integration.

Solution:



The X-axis is taken to coincide with the base.

A differential strip of area $dA = bdy$ is chosen.

Area of the rectangle $= bh$

Moment of area $= \int_0^h y.b\,dy$

$= b\left[\frac{y^2}{2}\right]_0^h$

$= b\left[\frac{h^2}{2}\right] = \frac{bh^2}{2}$

$\therefore\ \bar{y} = \frac{\text{Moment of area}}{\text{Total area}} = \frac{\frac{bh^2}{2}}{bh} = \frac{h}{2}$

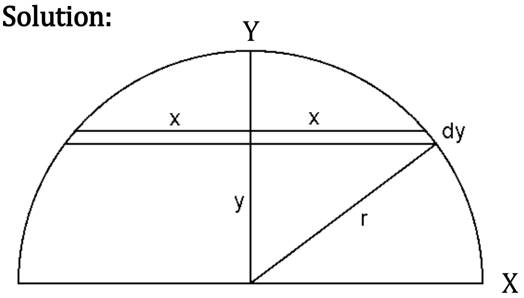
Thus the centroid of rectangle is at a distance $\frac{h}{2}$ from the base of the rectangle where h is the height of the rectangle.

Similarly, taking the strip parallel to Y-axis it can be shown that $\bar{x} = \frac{b}{2}$.

$\therefore$ The coordinate of centroid of rectangle, as shown in the figure above is $\left(\frac{b}{2}, \frac{h}{2}\right)$

29. Determine the centroid of a semi circle.

Solution:



Consider an infinitesimally small horizontal strip of thickness dy, at a distance y from the base, as shown in the figure. The length of the strip is $2x$.

By Pythagoras theorem, we get $x^2 + y^2 = r^2$

$$\Rightarrow x = \sqrt{r^2 - y^2}$$

The moment of all strips of the semicircle about the base $= \int_0^r 2xy\,dy$

$= 2\int_0^r y\sqrt{r^2 - y^2}\,dy$

$= -2\int_0^r z.z\,dz$

$= -2\int_0^r z^2\,dz$

$= -2\left[\frac{z^3}{3}\right]_0^r$

$= -2\left[\frac{\left(\sqrt{r^2-y^2}\right)^3}{3}\right]_0^r = -2\left[0 - \frac{r^3}{3}\right] = \frac{2r^3}{3}$

Let, $z = \sqrt{r^2 - y^2}$

$\Rightarrow z^2 = r^2 - y^2$

$\Rightarrow 2zdz = -2ydy$

$\Rightarrow -zdz = ydy$

$\therefore\ \bar{y} = \frac{\text{Moment of area}}{\text{Area of semicircle}} = \frac{\frac{2r^3}{3}}{\frac{\pi r^2}{2}} = \frac{4r}{3\pi}$

$\therefore$ The coordinates of centroid are $\left(0, \frac{4r}{3\pi}\right)$

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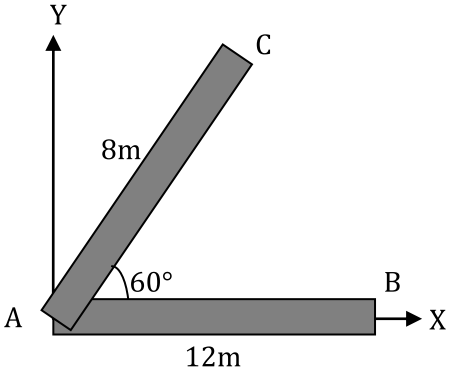
30. AB and AC are two uniform rods of lengths 12m and 8m respectively. If angle BAC=60°, find the distance from A to the CG of the two rods.

Solution:



Let, mass per unit length of rod is u.

Then mass of AB $= M_1 = 12u$

and mass of BC $= M_2 = 8u$

Let, CG of AB is at (x_1, y_1)

$\therefore x_1 = \frac{12}{2} = 6$ and $y_1 = 0$

Similarly, CG of AC is at (x_2, y_2)

$\therefore x_2 = \frac{8}{2}\cos 60° = 2$ and $y_1 = \frac{8}{2}\sin 60° = 2\sqrt{3}$

If CG of two rods to be $(\bar{x}, \bar{y})$ then

$\bar{x} = \frac{M_1x_1+M_2x_2}{M_1+M_2} = \frac{12u\times6+8u\times2}{12u+8u} = \frac{88u}{20u} = 4.4$

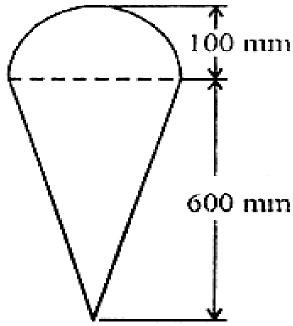
$\bar{y} = \frac{M_1y_1+M_2y_2}{M_1+M_2} = \frac{12u\times0+8u\times2\sqrt{3}}{12u+8u} = \frac{16\sqrt{3}u}{20u} = 1.38$

$\therefore$ Coordinate of CG of two rods $= (4.4, 1.38)$

$\therefore$ The distance from A to the CG of two rods

$= \sqrt{(4.4)^2 + (1.38)^2} = 4.61$ unit (Ans)

31. Locate the position of centroid of an ice-cream cone as shown in figure.



Solution: The given figure is symmetrical about Y-axis. Hence CG of section will lie on this axis. a_1 and a_2 are the area of triangle and semicircle respectively and y_1 and y_2 are their respective distance of CG from bottom line.

Now,

$a_1 = \frac{1}{2}\times 200 \times 600 = 60000\ mm^2$

$a_2 = \frac{\pi}{2}\times 100^2 = 15707.96\ mm^2$

$y_1 = \frac{2h}{3} = \frac{2\times600}{3} = 400\ mm$

$y_2 = 600 + \frac{4\times100}{3\pi} = 642.44\ mm$

$\therefore\ \bar{y} = \frac{a_1y_1+a_2y_2}{a_1+a_2}$

$= \frac{60000\times400+15707.96\times642.44}{60000+15707.96}$

$= 450.30\ mm$

Hence, centroid (G) for given ice cream cone lies at height of 450.30 mm from bottom.

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1. **An ideal machine is one whose efficiency is- a) 50% b) 50% to 70% c) <50% d) 100%.**
   Explanation: An ideal machine is one in which there is no energy loss due to friction or other resistances, meaning that the output work is equal to the input work. This results in 100% efficiency.
   Answer: (d)

2. **A lifting machine having an efficiency less than 50% is known as-**
   (a) reversible machine
   (b) non-reversible machine
   (c) ideal machine
   (d) none of these
   Explanation: The condition for a machine to be non-reversible or self-locking is that its efficiency should be less than 50%.
   Answer: (b)

3. **In actual machines**
   (a) MA=VR
   (b) MA>VR
   (c) MA<VR
   (d) MA=1
   Explanation: The term "actual machines" typically refers to physical machines or devices in the real world, as opposed to virtual machines or conceptual machines. In actual machine efficiency $\eta < 100\%$ or $\eta < 1$
   $\Rightarrow \frac{MA}{VR} < 1 , \Rightarrow MA < VR$
   Answer: (c)

4. **The velocity ratio of simple wheel and axle with 'D' and 'd' as the diameters of effort wheel and load axle, is-**
   (a) D+d
   (b) D−d
   (c) D×d
   (d) D/d
   Explanation:
   $$\text{Velocity Ratio} = \frac{\text{distance moved by the effort}}{\text{distance moved by the load}}$$
   $$= \frac{\pi D}{\pi d} = \frac{D}{d}$$
   Answer: (d)

5. **In simple lifting machine, Load vs Effort graph is-**
   (a) straight line
   (b) parabolic curve
   (c) hyperbolic curve
   (d) none of these
   Explanation: From law of machine the relation between load (W) and effort (P) is
   $$P = mW + c$$
   Here m and c are constant
   Thus it is an equation of straight line
   Answer: (a)

6. **The maximum mechanical advantage of a lifting machine is given by- a) 1+ m b) 1-m c) m/ VR d) 1/m.**
   Explanation: From the law of machine equation
   $$P = mW + c$$
   For maximum M.A. frictional resistance $c = 0$, the equation simplifies to- $P = mW$
   $\therefore M.A._{max} = \frac{W}{P} = \frac{W}{mW} = \frac{1}{m}$
   Answer: (d)

7. **Efficiency of a lifting machine is the ratio of- a) P/W b) V.R./M.A. c) M.A./V.R. d) M.A. x V.R.**
   Explanation: The efficiency (η) of a lifting machine is given by the ratio of Mechanical Advantage (M.A.) to Velocity Ratio (V.R.), expressed as:
   η = M.A./V.R.
   Answer: (c)

8. **The frictional load is given by-**
   (a) $\boldsymbol{P - \frac{W}{VR}}$
   (b) $\boldsymbol{P \times VR - W}$
   (c) $\boldsymbol{W - \frac{P}{VR}}$
   (d) none of these
   Explanation: In ideal condition $MA = VR$
   $$\Rightarrow \frac{W}{P} = VR$$
   $$\Rightarrow W = P \times VR$$
   $\therefore$ Ideal load $W_I = P \times VR$
   Frictional load $= W_I - W = P \times VR - W$
   Answer: (b)

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9. **The effort lost due to friction is given by-**
   (a) $P - \frac{W}{VR}$
   (b) $P \times VR - W$
   (c) $W - \frac{P}{VR}$
   (d) none of these
   Explanation: In ideal condition $MA = VR$
$$\Rightarrow \frac{W}{P} = VR$$
$$\Rightarrow P = \frac{W}{VR}$$
   $\therefore$ Ideal effort $P_I = \frac{W}{VR}$
   Effort lost due to friction $= P - P_I = P - \frac{W}{VR}$
   Answer: (a)

10. **The VR of single purchase crab winch is-**
   (a) $\frac{LN}{nr}$
   (b) $\frac{lT_1}{rT_2}$
   (c) $\frac{D}{D-d}$
   (d) $\frac{2\pi l}{p}$
   Explanation:
   $VR = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}} = \frac{2\pi l}{2\pi r \times \frac{T_2}{T_1}} = \frac{lT_1}{rT_2}$
   Notations are taken as usual.
   Answer: (b)

11. **In a simple lifting machine, the following parameters remains constant-**
   (a) Efficiency
   (b) Velocity ratio
   (c) Mechanical advantage
   (d) All of these
   Explanation: Mechanical advantage efficiency of a machine decreases due to the friction and weight of moving parts of the machine whereas the velocity ratio remains constant.
   Answer: (b)

12. **What is simple lifting machine?**
   Answer: The machines which have one point for the application of effort and one point for the load to be lifted are called simple lifting machines.
   Example: The simple lifting machine examples are Simple screw jack, lever, Nail cutter, Hand pump, etc.

13. **What are the advantages of a lifting machine?**
   Answer:
   The advantages of lifting machines include:
   1. Reduces Human Effort: Lifting machines make it easier to lift heavy objects, reducing physical strain on workers.
   2. Increases Efficiency: They allow tasks to be completed faster, improving productivity in industries like construction and manufacturing.
   3. Handles Heavy Loads: Machines can lift loads that would be impossible or extremely difficult for humans to lift manually.
   4. Improves Safety: Reducing manual lifting lowers the risk of workplace injuries such as back strain and muscle fatigue.
   5. Precision and Control: Many lifting machines have precise controls that allow accurate positioning of heavy objects.

14. **Define mechanical advantage.**
   Answer:
   Mechanical Advantage (MA): Mechanical advantage of a machine is defined as the ratio of load to the effort.
$$MA = \frac{\text{Load}}{\text{Effort}} = \frac{W}{P}$$
   Mechanical Advantage has no unit because it is the ratio of output force by input force.

15. **What do you mean by velocity ratio?**
   Answer:
   Velocity Ratio(VR): The ratio of the distance moved by the point at which the effort is applied in a simple machine to the distance moved by the point at which the load is applied, in the same time.
   In the case of an ideal machine,
   velocity ratio = mechanical advantage.

16. **Define efficiency of machine.**
   Answer:
   Efficiency: The efficiency of a machine is the ratio of the work done on the load by the machine to the work done on the machine by the effort. It is represented by symbol η.
   Formula of efficiency $\eta = \frac{\text{Output workdone}}{\text{Input workdone}}$

17. **Define input and output of a simple lifting machine.**
Solution:
Input of a Machine: The input of a machine is the work done on the machine. In a lifting machine, it is measured by the product of effort (P) and the distance (y) through which it has moved.
Input= (effort)×(distance moved by effort)
$= P \times y$
Output of a Machine: The output of a machine is the actual work done by the machine. In a lifting machine, it is measured by the product of the weight (W) lifted and the distance (x) through which it has been lifted.
Output= (load lifted)×(distance moved by load)
$= W \times x$

18. **Establish the relation between MA, VR and η.**
Solution:
Relation between MA, VR and Efficiency:
Let,
W = Load lifted by the machine,
P = Effort required to lift the load,
y = Distance moved by the effort in lifting the load
x = Distance moved by the load
By definition,
Mechanical Advantage $= \frac{\text{Load lifted by the machine}}{\text{Effort required to lift the load}}$
$\Rightarrow MA = \frac{W}{P}$
Velocity Ratio $= \frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$
$\Rightarrow VR = \frac{y}{x}$
Input =(effort)×(distance moved by effort)
⇒Input= $P \times y$
Output= (load lifted)×(distance moved by load)
⇒Output = $W \times x$
Now, efficiency $\eta = \frac{\text{Output}}{\text{Input}}$
$\Rightarrow \eta = \frac{W \times x}{P \times y}$
$\Rightarrow \eta = \frac{\frac{W}{P}}{\frac{y}{x}}$
$\Rightarrow \eta = \frac{MA}{VR}$
∴ Efficiency $= \frac{\text{Mechanical Advantage}}{\text{Velocity Ratio}}$

19. **What is an ideal machine?**
Answer:
Ideal Machine: A machine in which no part of the work done on the machine is wasted, is called an ideal machine or perfect machine. The efficiency of an ideal machine is 1 (or 100%) i.e., work output is equal to work input.

20. **What is reversible machine?**
Answer:
Reversible Machine: If a machine is capable of doing some work in the reversed direction, after the effort is removed, then the machine is known as reversible machine. The condition for a machine to be reversible is that its efficiency should be more than 50 %.

21. **Determine the condition of reversibility of a lifting machine.**
Solution:
Condition of reversibility: A machine is called as reversible, if it is able to do work in reverse direction after the removal of load from it.
For a actual machine,
Friction loss in machine = input – output
$= (P \times y) - (W \times x)$
For a reversible machine output of the machine is more than frictional loss.
∴ Output > Friction loss
$\Rightarrow W \times x > (P \times y) - (W \times x)$
$\Rightarrow 2(W \times x) > (P \times y)$
$\Rightarrow \frac{W \times x}{P \times y} > \frac{1}{2}$
$\Rightarrow \frac{\frac{W}{P}}{\frac{y}{x}} > \frac{1}{2}$
$\Rightarrow \frac{MA}{VR} > \frac{1}{2}, \Rightarrow \eta > \frac{1}{2}$ or 50%
Thus, for a reversible machine efficiency is greater than 50%.

22. **What is self-locking machine?**
Answer:
Self-Locking Machine: A machine is not capable of doing any work in the reversed direction, after the effort is removed. Such a machine is called a non-reversible or self-locking machine. The condition for a machine to be non-reversible or self-locking is that its efficiency should be less than 50%.

23. **Explain ideal load of a machine.**
Ideal Load: Ideal load is the load that can be lifted using the given effort by the machine, assuming it to be ideal. For the ideal machine, VR = MA
If W_I is the ideal load, then $W_I = P \times VR$.

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24. **Deduce the expression for the condition of self locking of a lifting machine.**

Solution:

Condition of self locking: A machine is called as self locking, if it is unable to do work in reverse direction after the removal of load from it.

For a actual machine,

Friction loss in machine = input – output

$= (P \times y) - (W \times x)$

For a self locking machine output of the machine is less than or equal to the frictional loss.

∴ Output ≤ Friction loss

$\Rightarrow W \times x \leq (P \times y) - (W \times x)$

$\Rightarrow 2(W \times x) \leq (P \times y)$

$\Rightarrow \frac{W \times x}{P \times y} \leq \frac{1}{2}$

$\Rightarrow \frac{\frac{W}{P}}{\frac{y}{x}} \leq \frac{1}{2}$

$\Rightarrow \frac{MA}{VR} \leq \frac{1}{2}, \Rightarrow \eta \leq \frac{1}{2}$ or 50%

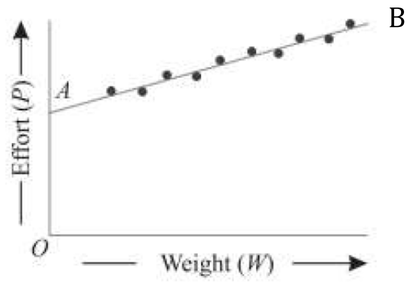
Thus for a self locking machine efficiency is less than or equal to 50%.

25. **State and explain law of machine.**

Answer:

Law of Machine: The equation which gives the relation between load (W) lifted and effort (P) applied in the form of a slope and intercept of a straight line is called as law of a machine. i.e.

$$P = mW + C$$



The intercept OA (or C) represents the amount of friction offered by the machine Or in other words, this is the effort required by the machine to overcome the friction, before it can lift any load. m =slope of AB it a constant called coefficient of friction.

26. **From equation of law of machine find the maximum mechanical advantage of a simple lifting machine.**

Solution:

If a simple lifting machine lifts a load of W by a effort of P then from law of machine we get,

$P = mW + c$ [where m and c are constant]

$\Rightarrow mW = P - c$

$\Rightarrow mW = P\left(1 - \frac{c}{P}\right)$

$\Rightarrow \frac{W}{P} = \frac{1}{m}\left(1 - \frac{c}{P}\right)$

$\Rightarrow MA = \frac{1}{m}\left(1 - \frac{c}{P}\right)$... (i) $\left[\because MA = \frac{W}{P}\right]$

Form equation (i) it is clear that with the decrease of $\frac{c}{P}$, MA increases. If the value of P so large, that the value of $\frac{c}{P}$ becomes negligible, then MA will be maximum.

∴ Maximum mechanical advantage, $MA_{max} = \frac{1}{m}$

27. **From equation of law of machine find the maximum efficiency of a simple lifting machine.**

Solution:

If a simple lifting machine lifts a load of W by a effort of P then from law of machine we get,

$P = mW + c$ [where m and c are constant]

$\Rightarrow mW = P - c$

$\Rightarrow mW = P\left(1 - \frac{c}{P}\right)$

$\Rightarrow \frac{W}{P} = \frac{1}{m}\left(1 - \frac{c}{P}\right)$

$\Rightarrow MA = \frac{1}{m}\left(1 - \frac{c}{P}\right)$ $\left[\because MA = \frac{W}{P}\right]$

$\Rightarrow \frac{MA}{VR} = \frac{1}{m \times VR}\left(1 - \frac{c}{P}\right)$

$\Rightarrow \eta = \frac{1}{m \times VR}\left(1 - \frac{c}{P}\right)$ ... (i) [η =Efficiency]

Form equation (i) it is clear that with the decrease of $\frac{c}{P}$, η increases. If the value of P so large, that the value of $\frac{c}{P}$ becomes negligible, then η will be maximum.

∴ Maximum efficiency, $\eta_{max} = \frac{1}{m \times VR}$

28. **If the law of lifting machine is $P = 0.04W + 20$, then find the maximum mechanical advantage of the machine.**

Solution:

Given law of machine $P = 0.04W + 20$

Comparing it with general law of machine

$$P = mW + c$$

We get $m = 0.04$

∴ Maximum MA $= \frac{1}{m} = \frac{1}{0.04} = 25$ (Ans)

29. **What do you mean by 'effort lost in friction' and 'frictional resistance'?**

Answer:

Effort Lost in Friction: When a machine or a mechanical system is in operation, some of the input effort (or energy) is wasted due to friction between moving parts. This wasted effort is called effort lost in friction. In ideal condition $MA = VR$

$$\Rightarrow \frac{W}{P} = VR$$

$$\Rightarrow P = \frac{W}{VR}$$

$\therefore$ Ideal effort $P_I = \frac{W}{VR}$

Effort lost due to friction $= P - P_I = P - \frac{W}{VR}$

Frictional Resistance: Frictional resistance (or simply friction) is the opposing force that resists the relative motion of two surfaces in contact. It depends on:

i) The normal reaction force (N) between the surfaces

ii) The coefficient of friction (μ)

Formula: F=μN

30. **The efficiency of a lifting machine is 80% when an effort 50N is required to lift a load of 1 kN. Find the mechanical advantage and velocity ratio of the machine.**

Solution:

Given data: Load $W = 1\text{ kN} = 1000\text{N}$

Effort $P = 50\text{N}$

Efficiency $\eta = 80\% = 0.8$

$\therefore$ Mechanical Advantage (MA) $= \frac{W}{P} = \frac{1000}{50} = 20$

We know efficiency $\eta = \frac{MA}{VR}$

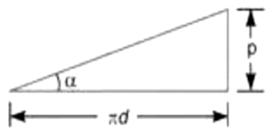
$\Rightarrow VR = \frac{MA}{\eta} = \frac{20}{0.8} = 25$

$\therefore$ Mechanical Advantage $= 20$ (Ans)

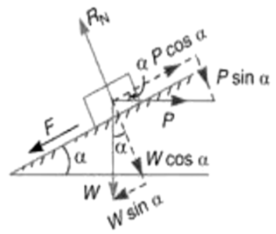
Velocity Ratio $= 25$ (Ans)

31. **What is the minimum effort P required to lift a load W by a screw jack when helix angle is α and coefficient of friction is $\tan\phi$ where ϕ is the angle of friction?**

Solution:



Development of screw



Forces acting on the screw

Let, $p =$ Pitch of the screw

$d =$ Mean diameter of the screw

$\mu =$ Coefficient of friction$= \tan\phi$

From the figure we find out that $\tan\alpha = \frac{p}{\pi d}$

Resolving the forces perpendicular to the plane

$R_N = P\sin\alpha + W\cos\alpha$

Resolving the forces along the plane

$P\cos\alpha = W\sin\alpha + F$

$= W\sin\alpha + \mu R_N$

$= W\sin\alpha + \mu(P\sin\alpha + W\cos\alpha)$

$\Rightarrow P\cos\alpha = W\sin\alpha + \mu P\sin\alpha + \mu W\cos\alpha$

$\Rightarrow P(\cos\alpha - \mu\sin\alpha) = W(\sin\alpha + \mu\cos\alpha)$

$\Rightarrow P = W \times \frac{(\sin\alpha + \mu\cos\alpha)}{(\cos\alpha - \mu\sin\alpha)}$

$\Rightarrow P = W \times \frac{(\sin\alpha + \tan\phi\cos\alpha)}{(\cos\alpha - \tan\phi\sin\alpha)}$ $[\because \mu = \tan\phi]$

$\Rightarrow P = W \times \frac{(\sin\alpha\cos\phi + \cos\alpha\sin\phi)}{(\cos\alpha\cos\phi - \sin\alpha\sin\phi)}$

$\Rightarrow P = W \times \frac{\sin(\alpha+\phi)}{\cos(\alpha+\phi)} = W\tan(\alpha+\phi)$

$\therefore$ The minimum effort required, $P = W\tan(\alpha+\phi)$

32. **In a differential wheel and axle the larger axle is 200mm in diameter and the smaller one is 150mm in diameter. The effort handle is 400mm long. Find the velocity ratio. If an effort of 150N lifts a load of 1500N, find the efficiency.**

Solution:

Given, Effort handle $L = 400mm$

Large axle diameter $d_1 = 200mm$

Small axle diameter $d_2 = 150mm$

Load lifted $W = 1500\text{N}$

Effort required $P = 150\text{N}$

Diameter of effort wheel $D = 2L$

$= 2 \times 400$

$= 800mm$

$\therefore$ Velocity Ratio $VR = \frac{2D}{d_1 - d_2} = \frac{2\times 800}{200-150} = 32$

Mechanical Advantage $MA = \frac{W}{P} = \frac{1500}{150} = 10$

Efficiency $\eta = \frac{MA}{VR} = \frac{10}{32} = 0.3125 = 31.25\%$

33. **In a differential wheel and axle, the diameter of the effort wheel is 40cm. The radii of axles are 15cm and 10cm respectively. Find the load which can be lifted by an effort of 100N, assuming the efficiency of the machine to be 75%.**

Solution:

Given, Diameter of effort wheel $D = 40cm = 0.4m$

The radius of axles, 15cm and 10cm

∴Diameter of axles are

$d_1 = 2 \times 15 = 30cm = 0.3m$

$d_2 = 2 \times 10 = 20cm = 0.2m$

∴ Velocity ratio $VR = \frac{2D}{d_1 - d_2} = \frac{2\times0.4}{0.3-0.4} = 8$

Effort required $P = 100\text{N}$

Efficiency $\eta = 75\% = 0.75$

We know, $\eta = \frac{MA}{VR}$

$\Rightarrow 0.75 = \frac{W/P}{8}$

$\Rightarrow \frac{W}{P} = 6$, $\Rightarrow W = 6P = 6 \times 100 = 600\text{N}$

34. **The no of teeth on the worm wheel of a double threaded worm and worm wheel is 25. The diameter of effort wheel is 25cm and load drum is of 15cm diameter. Find the efficiency of the machine, if an effort of 300N can lift load of 3.25kN.**

Solution:

Given, no of thread $n = 2$ [For double threaded]

No of teeth on worm wheel $T = 25$

Diameter of load drum $= 15cm$

Radius of load drum $r = \frac{15}{2} = 7.5cm$

Diameter of effort wheel $D = 25cm$

Effort required $P = 300N$

Load lifted $W = 3.25\text{kN}=3250\text{N}$

∴ Velocity ratio $VR = \frac{DT}{2nr} = \frac{25\times25}{2\times2\times7.5} = 20.83$

Mechanical advantage $MA = \frac{W}{P} = \frac{3250}{300} = 10.83$

Efficiency $\eta = \frac{MA}{VR} = \frac{10.83}{20.83} = 52\%$

35. **The pitch of a screw of a jack is 10mm and the mean diameter of the thread is 60mm and the length of lever is 500mm. Find effort required to lift a load of 10kN. Take coefficient of friction $\mu = 0.08$.**

Solution:

Given, pitch of screw $p = 10mm$

Diameter of thread $d = 60mm$

Load lifted $W = 10\text{kN}=10\times10^3\text{N}$

Coefficient of friction $\mu = 0.08$

$\Rightarrow \tan\phi = 0.08$

If α =Helix angle then $\tan\alpha = \frac{p}{\pi d} = \frac{10}{\pi\times60} = 0.053$

∴ Effort required to lift the load,

$P = W\tan(\alpha + \phi)$

$= 10 \times 10^3 \times \frac{\tan\alpha + \tan\phi}{1 - \tan\alpha . \tan\phi}$

$= 10 \times 10^3 \times \frac{0.053+0.08}{1-0.053\times0.08}$

$= 10 \times 10^3 \times \frac{0.133}{0.9576} = 1388.89\text{N}$ (Ans)

36. **A simple screw jack of has a thread of pitch of 10mm, find mechanical advantage and efficiency, if an effort of 20N is applied at the end of an arm 500mm long to lift a load 2.6kN**

Solution: Given,

Pitch of screw $p = 10mm$

Length of arm $l = 500mm$

Effort required $P = 20\text{N}$

Load lifted $W = 2.6\text{kN}=2600\text{N}$

∴ Velocity Ratio $= \frac{2\pi l}{p} = \frac{2\pi\times500}{10} = 314.16$

Mechanical Advantage $= \frac{W}{P} = \frac{2600}{20} = 130$

∴ The efficiency of the screw jack

$= \frac{MA}{VR} = \frac{130}{314.16} = 0.4138 = 41.38\%$

37. **Explain ideal effort of a machine.**

Answer:

Ideal Effort: Ideal effort is the effort required to lift the given load by the machine assuming the machine to be ideal. For ideal machine, $VR = MA$

If P_I is the ideal effort, then $VR = \frac{W}{P_I}$

Hence, $P_I = \frac{W}{VR}$

38. **To lift a load of 1500N, an effort of 200N is applied to a machine. The load is lifted by a distance of 0.2m while the effort is moved by a distance of 2m. Find the VR, MA and efficiency of the machine. Find also the ideal effort, effort lost due to friction, ideal load and frictional load.**

Solution:

Given, Load lifted $W = 1500\text{N}$

Effort required $P = 200\text{N}$

Distance moved by load $x = 0.2m$

Distance moved by effort $y = 2m$

$\therefore$ Velocity Ratio $VR = \frac{y}{x} = \frac{2}{0.2} = 10$

Mechanical Advantage $MA = \frac{W}{P} = \frac{1500}{200} = 7.5$

Efficiency $\eta = \frac{MA}{VR} = \frac{7.5}{10} = 0.75 = 75\%$

Ideal effort $P_I = \frac{W}{VR} = \frac{1500}{10} = 150\text{N}$

Ideal load $W_I = P \times VR = 200 \times 10 = 2000\text{N}$

Effort lost due to friction $(P - P_I)$

$= 200 - 150 = 50\text{N}$

Frictional load $W_I - W = 2000 - 1500 = 500\text{N}$

39. **In a Weston's differential pulley block, the difference in the number of teeth of the two pulleys is 4. If the efficiency of the machine is 50% and an effort of 100N just lift a load of 2kN, find the number of teeth of the two pulleys.**

Solution:

Given, Load lifted $W = 2\text{kN} = 2000\text{N}$

Effort required $P = 100\text{N}$

Efficiency $\eta = 50\% = 0.5$

Difference in no of teeth $(T_1 - T_2) = 4$

$\therefore$ Mechanical Advantage $MA = \frac{W}{P} = \frac{2000}{100} = 20$

We know, $\eta = \frac{MA}{VR}$

$\Rightarrow VR = \frac{MA}{\eta} = \frac{20}{0.5} = 40$

Now, for Weston's differential pulley block

$VR = \frac{2T_1}{T_1 - T_2}$

$\Rightarrow 40 = \frac{2T_1}{4}, \Rightarrow T_1 = 40 \times \frac{4}{2} = 80$ (Ans)

$\therefore T_2 = 80 - 4 = 76$ (Ans)

40. **In a simple lifting machine, efforts of magnitude 0.4kN and 0.7kN are required to be applied to lift loads of magnitude 6kN and 12kN respectively. The velocity ratio of the machine is 32. Find:**
(i) The law of the machine.
(ii) Efficiencies of the machine at both the load options.
(iii) Effort lost in friction in both the cases.
(iv) Maximum MA and maximum efficiency.
Also give your opinion on the reversibility of the machine in both load effort combination.

Solution:

(i)Let the law of machine, $P = mW + c$

From the question if $P = 0.4\text{kN}$, then $W = 6\text{kN}$

$\therefore 0.4 = m \times 6 + c$

$\Rightarrow 6m + c = 0.4$ .... (1)

Similarly, if $P = 0.7\text{kN}$, then $W = 12\text{kN}$

$\therefore 0.7 = m \times 12 + c$

$\Rightarrow 12m + c = 0.7$ .... (2)

By (2) – (1) we get,

$12m - 6m = 0.7 - 0.4$

$\Rightarrow 6m = 0.3, \Rightarrow m = \frac{0.3}{6} = 0.05$

From equation (1) we get,

$6 \times 0.05 + c = 0.4, \Rightarrow c = 0.4 - 0.3 = 0.1$

$\therefore$ The law of the machine is $P = 0.05W + 0.1$

(ii) Given that $VR = 32$

For load of 6kN, Effort required 0.4kN

$\therefore$ Efficiency $\eta_1 = \frac{MA}{VR} = \frac{\frac{6}{0.4}}{32} = 0.4687 = 46.87\%$

For load of 12kN, Effort required 0.7kN

$\therefore$ Efficiency $\eta_2 = \frac{MA}{VR} = \frac{\frac{12}{0.7}}{32} = 0.5357 = 53.57\%$

(iii) Effort lost in friction given by

$P_F = P - \frac{W}{VR}$

For 6kN load, $P_F = 0.4 - \frac{6}{32} = 0.2125\text{N}$

For 12kN load, $P_F = 0.7 - \frac{12}{32} = 0.325\text{N}$

(iv) Maximum MA $= \frac{1}{m} = \frac{1}{0.05} = 20$

Maximum efficiency $= \frac{1}{m \times VR}$

$= \frac{1}{0.05 \times 32} = 0.625 = 62.5\%$

Since, the efficiency of the machine for 6kN load is 46.87% (<50%) therefore the machine is irreversible of self locking in 1st case. But the efficiency of the machine for 12kN load is 53.56% (>50%) so the machine is reversible in 2nd case.

41. **The law of machine, with a velocity ratio 160, is given by $P = 0.014W + 15$.**
Determine:
(a) The effort required to lift a load of 50kN
(b) Mechanical advantage under condition(a)
(c) Efficiency under condition (a)
(d) Maximum efficiency of the machine
Solution:
(a) Load lifted $W = 50\text{kN} = 50 \times 10^3\text{N}$
Given law of machine $P = 0.014W + 15$.
$\therefore$ Effort required $P = 0.014 \times 50 \times 10^3 + 15$
$= 715\text{N}$ (Ans)
(b) Mechanical Advantage $MA = \frac{W}{P}$
$= \frac{50\times10^3}{715}$
$= 69.93$ (Ans)
(c) Given velocity ratio $VR = 160$
$\therefore$ Efficiency $\eta = \frac{MA}{VR}$
$= \frac{69.93}{160} = 0.437 = 43.7\%$ (Ans)
(d) Maximum efficiency
$\eta_{max} = \frac{1}{m\times VR}$
$= \frac{1}{0.014\times160} = 0.4464 = 44.64\%$ (Ans)

42. **What load will be lifted by an effort of 12 N, if the velocity ratio is 18 and efficiency of the machine at this load is 60%? If the machine has a constant friction resistance, determine the law of the machine.**
Solution: Given, $P = 12\text{ N}, VR = 18, \eta = 60\% = 0.6$
Now, $\frac{MA}{VR} = \eta, \Rightarrow MA = \eta \times VR = 0.6 \times 18 = 10.8$
And, $\frac{W}{P} = MA$
$\Rightarrow W = MA \times P = 10.8 \times 12 = 129.6\text{ N}$ (Ans)
$m = \frac{1}{VR} = \frac{1}{18} = 0.06$
Replacing the values on $P = mW + c$
$\Rightarrow 12 = 0.06 \times 129.6 + c$
$\Rightarrow c = 12 - 7.78 = 4.22$
$\therefore$ The law of machine is $P = 0.06W + 4.22$ (Ans)

43. **The law of wheel and axle machine is $P = 0.2W + 50$. The diameter of wheel is 600 mm and radius of axle is 50 mm. Determine-**
(a) Mechanical Advantage,
(b) Effort required to lift a load of 2000N,
(c) Maximum Efficiency,
(d) Maximum Mechanical Advantage and
(e) Load to be lifted by an effort of 520 N.
Solution: Given, Diameter of wheel $D = 600\ mm$
Radius of axle $r = 50\ mm, \therefore d = 2 \times 50 = 100\ mm$
$\therefore$ VR of the machine $= \frac{D}{d} = \frac{600}{100} = 6$
(a) Mechanical Advantage according to condition (b)
$MA = \frac{W}{P} = \frac{2000}{450} = 4.44$ (Ans)
(b) Given load $W = 2000$ N
$\therefore$ Effort required $P = 0.2W + 50$
$\Rightarrow P = 0.2 \times 2000 + 50 = 450$ N (Ans)
(c) Maximum efficiency
$$\eta_{max} = \frac{1}{m \times VR} \times 100\%$$
$= \frac{1}{0.2\times6} \times 100\% = 83.33\%$ (Ans)
(d) Maximum mechanical advantage
$MA_{max} = \frac{1}{m} = \frac{1}{0.2} = 5$ (Ans)
(e) Given effort $P = 520$ N
$P = 0.2W + 50$
$\Rightarrow 520 = 0.2W + 50$
$\Rightarrow 0.2W = 520 - 50 = 470$
$\Rightarrow W = \frac{470}{0.2} = 2350$ N (Ans)

44. **In a double purchase crab winch, no of teeth on pinion are 20 and 30 and that of spur gears are 60 and 80. Radius of the load drum is 200mm and length of the handle is 300mm. Find the efficiency of the machine, if the effort 100N is required to lift a load of 750N.**
Solution:
Given, Length of handle $l = 300mm$
Radius of load drum $r = 200mm$
No of teeth on pinion $T_1 = 60$ ans $T_3 = 80$
No of teeth on spur gears $T_2 = 20$ ans $T_4 = 30$
Effort required $P = 100\text{N}$
Load lifted $W = 750\text{N}$
$\therefore$ Velocity Ratio (VR) $= \frac{l}{r} \times \frac{T_1}{T_2} \times \frac{T_3}{T_4}$
$= \frac{300}{200} \times \frac{60}{20} \times \frac{80}{30} = 12$
Mechanical Advantage (MA) $= \frac{W}{P} = \frac{750}{100} = 7.5$
$\therefore$ Efficiency $\eta = \frac{MA}{VR} = \frac{7.5}{12} = 0.625 = 62.5\%$ (Ans)

{Prepared by NatiTute YouTube Channel}

45. **Explain the working principle of screw jack.**

Solution:

Screw Jack: It consist of a screw, fitted in a nut, which forms the body of the jack.

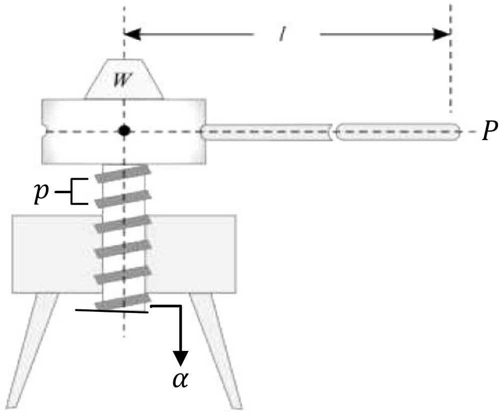


Figure shows a simple screw jack, which is rotated by the application of an effort at the end of the lever, for lifting the load. Now consider a single threaded simple screw jack

l = Length of the effort arm

p = Pitch of the screw

W = Load lifted by the screw jack

P = Effort applied at the end of the arm

∴ Distance moved by the effort in on revolution

$$y = 2\pi l$$

and distance moved by the load $x = p$

∴ Velocity Ration $= \frac{\text{Distance moved by th effort}}{\text{Distance moved by the load}}$

$$\Rightarrow VR = \frac{y}{x}$$

$$\Rightarrow VR = \frac{2\pi l}{p}$$

Mechanical Advantage $= \frac{\text{Load lifted by the screw jack}}{\text{Effort applied on screw jack}}$

$$\Rightarrow MA = \frac{W}{P}$$

Efficiency $\eta = \frac{MA}{VR}$

Note: The value of P i.e. the effort applied may also found by the relation: $P = W\tan(\alpha + \phi)$

Where, α= Helix angle

ϕ =Coefficient of friction

46. **Write the expression of VR of a simple screw jack.**

Answer:

Consider a single threaded simple screw jack

l = Length of the effort arm

p = Pitch of the screw

∴ Distance moved by the effort in on revolution

$$= 2\pi l$$

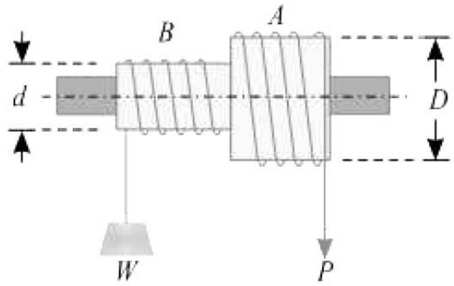
and distance moved by the load $= p$

∴ Velocity Ratio (VR) $= \frac{\text{Distance moved by th effort}}{\text{Distance moved by the load}}$

$$\Rightarrow VR = \frac{2\pi l}{p}$$

47. **With usual notation determine the VR and MA of simple wheel and axle.**

Solution:



In figure is shown a simple wheel and axle. A string is wound round the axle B, which carries the load to be lifted. A second string is wound round the wheel A in the opposite direction to that of the string on B.

Let,

D = Diameter of effort wheel

d = Diameter of the load axle

W = Load lifted

P = Effort applied to lift the load.

Velocity Ratio $= \frac{\text{Distance moved by th effort}}{\text{Distance moved by the load}}$

$$\Rightarrow VR = \frac{\pi D}{\pi d} = \frac{D}{d}$$

Mechanical Advantage $= \frac{\text{Load lifted by the screw jack}}{\text{Effort applied on screw jack}}$

$$\Rightarrow MA = \frac{W}{P}$$

{Prepared by NatiTute YouTube Channel}

1. **Newton's first law of motion gives the concept of –**
   (a) work
   (b) force
   (c) inertia
   (d) energy
   Explanation: Newton's first law of motion defines the inertia of the body. It states that everybody has a tendency to remain in its state (either rest or motion) due to its inertia.
   Answer: (c)

2. **Newton's which law is known as inertia law- a) first law b) second law c) third law d) fourth law**
   Explanation: Newton's first law of motion is known as inertia law, because it defines the inertia of the body. It states that everybody has a tendency to remain in its state (either rest or motion) due to its inertia.
   Answer: (a)

3. **A jet engine works on the principle of conservation of –**
   (a) mass
   (b) energy
   (c) linear momentum
   (d) angular momentum
   Explanation: The Jet engine works on the phenomenon of Conservation of Linear Momentum. It produces a large volume of gases through the combustion of fuel, which is allowed to escape in backward direction through a jet. The backward rushing gas gains large momentum because of high velocity.
   Answer: (c)

4. **The angular velocity of the second hand of a clock is-**
   (a) 2π rad/s
   (b) $\frac{\pi}{3}$ rad/s
   (c) $\frac{\pi}{30}$ rad/s
   (d) $\frac{\pi}{1800}$ rad/s
   Explanation: The Seconds hand of the clock takes 60 seconds to cover one cycle of 2π radian
   ∴ Angular velocity $\omega = \frac{\theta}{t} = \frac{2\pi}{60} = \frac{\pi}{30}$ rad/s
   Answer: (c)

5. **In a velocity-time graph, area enclosed with the time axis represents-**
   (a) displacement
   (b) velocity
   (c) force
   (d) distance
   Explanation:
   Area enclosed under velocity-time graph
   $= \text{Velocity} \times \text{time}$
   $= \frac{\text{displacement}}{\text{time}} \times \text{time} = \text{displacement}.$
   Answer: (a)

6. **The distance travelled by a body moving with an uniform acceleration in the n^{th} second is equal to- a) $2u + a(2n-1)$ b) $u + \frac{a}{2}(2n-1)$ c) $un + \frac{1}{2}an^2$ d) none of these.**
   Explanation: The formula for the distance traveled in the n^{th} second by a body moving with uniform acceleration is given by: $S_n = u + \frac{a}{2}(2n-1)$
   Answer: (b)

7. **For maximum range of a projectile, the angle of projection should be- a) 90° b) 60°c) 30° d) 45°.**
   Explanation: The range of a projectile is given by the formula: $R = \frac{u^2 \sin 2\theta}{g}$. The range is maximum when $\sin 2\theta$ is maximum. The maximum value of $\sin 2\theta$ is 1.
   $\therefore \sin 2\theta = 1, \Rightarrow \sin 2\theta = \sin 90°, \Rightarrow 2\theta = 90°,$
   $\Rightarrow \theta = 45°$
   Answer: (d)

8. **The energy possessed by a body due to its change in position or shape is called:**
   (a) Potential Energy
   (b) Kinetic Energy
   (c) Conservative Energy
   (d) Electrostatic Energy
   Explanation: The energy possessed by a body by virtue of its position is called the potential energy. Therefore, a body placed at a height above the ground possesses potential energy with respect to the ground.
   Answer: (a)

{Prepared by NatiTute YouTube Channel}

9. A particle is projected at an angle 45° with the horizontal with velocity of projection 150 m/s. Determine-
i) the maximum height attained by the projectile,
ii) the horizontal range of projectile,
iii) time of flight.

Solution: Given, angle of projection $\alpha = 45°$
Velocity of projection $u = 150$ m/s
Therefore,
i) The maximum height attained by the projectile
$= \frac{u^2 \sin^2 \alpha}{2g}$ [With usual notation]
$= \frac{150^2 \times \sin^2 45°}{2\times 9.81} = 573.39\, m$ (Ans)
ii) the horizontal range of projectile,
$= \frac{u^2 \sin 2\alpha}{g} = \frac{150^2 \sin(2\times 45°)}{9.81} = 2293.58\, m$ (Ans)
iii) Time of flight
$= \frac{2u \sin \alpha}{g} = \frac{2\times 150 \times \sin 45°}{9.81} = 21.62$ second (Ans)

10. Write the unit of angular velocity.

Answer: Angular velocity is represented by the Greek letter omega (ω, sometimes Ω). It is measured in angle per unit time; hence, the SI unit of angular velocity is radians per second or rad/s.

11. Find the final velocity of the stone, falling freely from a height of 5m to the ground.

Solution:
We know, for a freely falling body
$v^2 = u^2 + 2gh$ [With usual notation]
Here, $u = 0, h = 5m, v = ?$, taking $g = 9.8\, ms^{-2}$
$\therefore v^2 = 0 + 2 \times 9.8 \times 5$
$\Rightarrow v = \sqrt{2 \times 9.8 \times 5} = 9.90\, ms^{-1}$ (Ans)

12. Establish the relation between linear velocity and angular velocity.

Relation between linear velocity and angular velocity: The distance s covered by a body travelling in a circular path of radius r and turning its radial line by θ is given by-

$$s = r\theta$$

Differentiating both side with respect to time (t) we have
$\frac{ds}{dt} = r\frac{d\theta}{dt}$
i.e. $v = r\omega$ [With usual notation]
or, Linear velocity = Radius × Angular velocity

13. A man lifts a suitcase of 5kg mass from ground level to a 10.193m high rooftop. Find change in potential energy in that suitcase.

Solution:
Given, mass of suitcase $m = 5$kg
Height lifted $h = 10.193$m
∴ Potential energy stored =mgh
$= 5 \times 9.8 \times 10.193$
$= 499.46$ Jule (Ans)

14. Write the law of conservation of momentum.

Answer:
Law of conservation of momentum: Law of conservation of momentum states that. For two or more bodies in an isolated system acting upon each other, their total momentum remains constant unless an external force is applied. Therefore, momentum can neither be created nor destroyed.

15. What is the centripetal force?

Answer:
Centripetal Force: When an object which moves in a circular path experiences a force acting radially inwards called centripetal force. The force is directed inward toward the rotational axis.
The mathematical form of centripetal force is
$F = \frac{mv^2}{r}$ [With usual notation]
The centripetal force is observed from an inertial frame of reference.

16. A ball is thrown vertically upwards from ground, with an initial velocity of 20m/s. Find the greatest height attained by the ball.

Solution:
Given, initial velocity $u = 20$ m/s
Now, $v^2 = u^2 - 2gh$ [−ve sign for retardation]
$\Rightarrow 0 = 20^2 - 2 \times 9.8 \times h$
$\Rightarrow 19.6h = 400$
$\Rightarrow h = \frac{400}{19.6} = 20.41m$
∴ Greatest height attained by the ball is 20.41m.

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17. What is centrifugal force?

Answer:

Centrifugal Force: Centrifugal force is a pseudo force in a circular motion which acts along the radius and is directed away from the centre of the circle. The force does not exist when measurements are made in an inertial frame of reference. It only comes into play when changing our reference frame from a ground/inertial to a rotating reference frame.

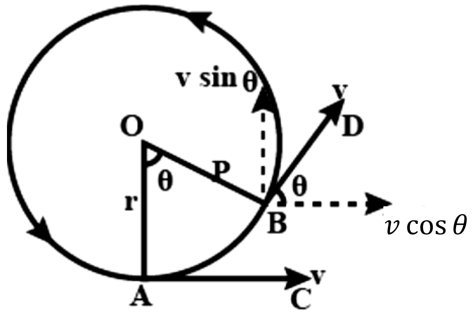
The mathematical form of centrifugal force is opposite to centripetal force i.e.

$F = -\frac{mv^2}{r}$ [With usual notation]

18. Deduce an expression for centripetal force.

Solution:

Let a particle of mass m be moving around the circle of radius r with a uniform speed v.



Let the particle moves from A to B in a time t sec. covering a small angle θ. At the point B velocity of the particle is along BD. This can be resolved into two components $v\cos\theta$ and $v\sin\theta$. Since θ is small $\sin\theta \cong \theta$ and $\cos\theta \cong 1$

∴ Change in velocity parallel to AC

$= v\cos\theta - v = v - v = 0$

Change in velocity parallel to OA

$= v\sin\theta - 0 = v\theta$

∴ Acceleration along OA $= \frac{v\theta}{t} = v\omega$

So, Centripetal force = Mass × Acceleration

$\Rightarrow F = mv\omega$

$\Rightarrow F = m\omega^2 r \quad [\because v = \omega r]$

$\Rightarrow F = \frac{mv^2}{r}$

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19. A train was moving at a constant velocity of 90km/h, as it approached a station, the loco-pilot applied brakes at a constant retardation of 0.5m/s² and stopped it at the station. How much time will it take to stop?

Solution:

Given,

Initial Velocity, $u = 90kmph = 90 \times \frac{5}{18} = 25ms^{-1}$

Retardation, $a = -0.5\, ms^{-2}$

Final velocity, $v = 0$

Let time taken $= t$ sec

Now, $v = u + at$

$\Rightarrow 0 = 25 + (-0.5) \times t$

$\Rightarrow 0.5t = 25\,, \Rightarrow t = \frac{25}{0.5} = 50$ sec (Ans)

20. **A wheel rotating about a fixed axis at 20 revolutions per minute is uniformly accelerated for 70 sec. during which it makes 50 revolutions. Find the-**

(i) angular velocity at the end of this interval

(ii) time required for the velocity to reach 100 revolution per minute.

Solution:

Given

Initial angular velocity $\omega_1 = 20$ rpm

$= 20 \times \frac{2\pi}{60}$ rad/s

$= \frac{2\pi}{3}$ rad/s

Angular displacement $\theta = 50$ revolution

$= 50 \times 2\pi$ rad

$= 100\pi$ rad

Time taken $t = 70$ sec

Now, $\theta = \omega_1 t + \frac{1}{2}\alpha t^2$ [α =Angular acceleration]

$\Rightarrow 100\pi = \frac{2\pi}{3} \times 70 + \frac{1}{2}\alpha \times 70^2$

$\Rightarrow 2450\alpha = 100\pi - \frac{140\pi}{3} = \frac{160\pi}{3}$

$\Rightarrow \alpha = \frac{160\pi}{3\times 2450} = 0.068\, rad/s^2$

(i) Angular velocity at 70 sec,

$\omega_2 = \omega_1 + \alpha t$

$= \frac{2\pi}{3} + 0.068 \times 70 = 6.85$ rad/s (Ans)

(ii) If final angular velocity

$\omega_2 = 100$ rpm$=100 \times \frac{2\pi}{60} = \frac{10\pi}{3}$ rad/s

Let time taken $= t$ sec

then, $\omega_2 = \omega_1 + \alpha t$

$\Rightarrow \frac{10\pi}{3} = \frac{2\pi}{3} + 0.068t$

$\Rightarrow 0.068t = \frac{8\pi}{3}$

$\Rightarrow t = \frac{8\pi}{3\times 0.068} = 123.20$ sec. (Ans)

21. Deduce the equation $v = u + at$.

Solution:

We know that the rate of change of velocity is the definition of body acceleration. Let us assume a body that has a mass "m" and initial velocity "u". Let after time "t" its final velocity becomes "v" due to uniform acceleration "a".

Now we know that:

$\text{Acceleration} = \frac{\text{Final Velocity} - \text{Initial Velocity}}{\text{Time Taken}}$

$\Rightarrow a = \frac{v-u}{t}$

$\Rightarrow at = v - u$

$\Rightarrow v = u + at$ (Proved)

22. A train was moving at a constant velocity of 80 km/h, as it approached a station, the loco-pilot applied brakes at a constant retardation of 0.74 m/sec^2 and stopped it at the station. How much time will it take to stop?

Solution:

Given,

Initial velocity $u = 80\frac{km}{h} = 80 \times \frac{5}{18} = 22..22\, m/s$

Final velocity $v = 0$

Retardation or -ve acceleration $a = -0.74\, m/s^2$

Using equation of motion $v = u + at$

We can say, $0 = 22.22 + (-0.74) \times t$

$\Rightarrow 0.74t = 22.22$

$\Rightarrow t = \frac{22.22}{0.74} = 30.03$ sec.

$\therefore$ The train will take 30.03 to stop.

23. Deduce the equation $s = ut + \frac{1}{2}at^2$

Solution:

Let the distance be "s".

Distance = Average velocity × Time.

And Average velocity $= \frac{u+v}{2}$

$\therefore$ Distance $s = \frac{u+v}{2} \times t$

From the first equation of motion, we know that $v = u + at$. Putting this value of v in the above equation, we get

$s = \frac{(u+(u+at))}{2} \times t$

$\Rightarrow s = \frac{2u+at}{2} \times t$

$\Rightarrow s = \left(\frac{2u}{2} + \frac{at}{2}\right) \times t$

$\Rightarrow s = ut + \frac{1}{2}at^2$ (Proved)

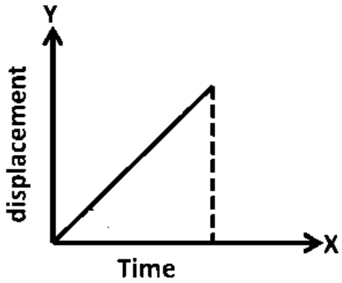
24. **Draw displacement-time graph**

(i) when velocity is uniform

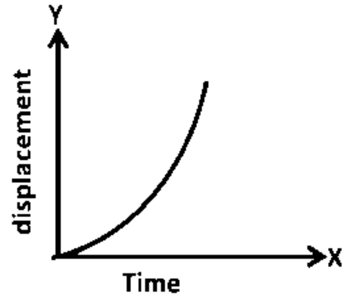
(ii) when velocity increasing

Answer:

(i) Displacement versus time graph when velocity is uniform, is drawn below



(ii) Displacement versus time graph when velocity is increasing, is drawn below



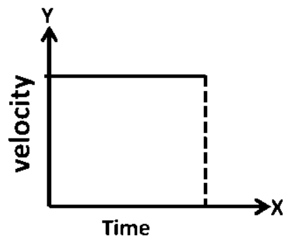
25. **Draw velocity-time graph**

(i) when velocity is uniform

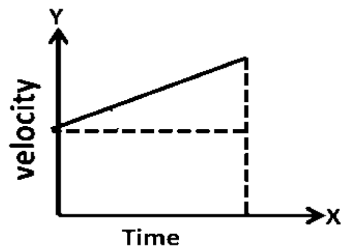
(ii) when velocity increasing starting with an initial velocity

Answer:

(i) Velocity versus time graph when velocity is uniform, is drawn below



(ii) Velocity versus time graph when velocity is uniform starting with an initial velocity is drawn below



26. Deduce the equation $v^2 = u^2 + 2as$

Solution:

We know that displacement is the product of average velocity and time. Mathematically, this can be represented as $s = \left(\frac{u+v}{2}\right) \times t$

From the first equation of motion, we know that

$$v = u + at$$

Rearranging the above formula, we get

$$t = \frac{v-u}{a}$$

Substituting the value of t in the displacement formula, we get

$s = \left(\frac{u+v}{2}\right) \times \frac{v-u}{a}$

$\Rightarrow s = \frac{v^2-u^2}{2a}$

$\Rightarrow v^2 - u^2 = 2as$

$\Rightarrow v^2 = u^2 + 2as$ (Proved)

27. Derive a relationship for the distance travelled by a body in the n^{th} second.

Answer: The velocity at any time t is given by-

$$v = u + at$$

Velocity after $(n-1)$ seconds,

$$v_1 = u + a(n-1)$$

Velocity after n seconds,

$$v_2 = u + an$$

Since the motion is under uniform acceleration, we can approximate the displacement (S_n) using the average velocity in this interval:

$S_n = v_{avg} \times Time$

$S_n = \frac{v_1+v_2}{2} \times 1$ [Since the time interval is 1 second]

$\Rightarrow S_n = \frac{u+a(n-1)+u+an}{2} \times 1$

$\Rightarrow S_n = \frac{2u+2an-a}{2}$

$\Rightarrow S_n = u + a\left(\frac{2n-1}{2}\right)$

$\Rightarrow S_n = u + a\left(n - \frac{1}{2}\right)$

28. Define angular acceleration.

Answer:

Angular Acceleration: Angular Acceleration is defined as the time rate of change of angular velocity. It is usually expressed in radians per second per second or rad/s^2. It is denoted by α

$$\alpha = \frac{d\omega}{dt}$$

The angular acceleration is also known as rotational acceleration. It is a quantitative expression of the change in angular velocity per unit time.

29. State the Newton's law of motion. Deduce the relationship P = mf.

Answer:

Newton's Laws of Motion

First Law (Law of Inertia): A body continues to remain in its state of rest or uniform motion in a straight line unless acted upon by an external force.

Second Law (Law of Acceleration): The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force.

Third Law (Action-Reaction Law): For every action, there is an equal and opposite reaction.

Derivation of $F = ma$: Newton's second law of motion states that the force applied to the system is equal to the time rate of change of momentum.

So, $F = \frac{dP}{dt}$ [Notations are taken as usual]

But, $P = mv$

So, $F = \frac{d(mv)}{dt}$

$\Rightarrow F = m\frac{dv}{dt}$ [$\because m$ is constant]

$\Rightarrow F = ma$ (Proved)

30. From Newton's second law of motion prove that $F = ma$.

Solution:

Newton's second law of motion states that the force applied to the system is equal to the time rate of change of momentum.

So, $F = \frac{dP}{dt}$ [Notations are taken as usual]

But, $P = mv$

So, $F = \frac{d(mv)}{dt}$

$\Rightarrow F = m\frac{dv}{dt}$ [$\because m$ is constant]

$\Rightarrow F = ma$ (Proved)

31. State the unit of angular velocity, angular acceleration and angular displacement.

Answer:

Physical Quantity	SI Unit
Angular Velocity	radian per second (rad/s)
Angular Acceleration	radian per second² (rad/s²)
Angular Displacement	radian (rad)

32. Establish the relation between linear acceleration and angular acceleration.

Relation between linear acceleration and angular acceleration: Angular acceleration is the rate of change of angular velocity

$$\alpha = \frac{d\omega}{dt}$$

Linear acceleration is rate of change of linear velocity

$$a = \frac{dv}{dt}$$

We know $v = \omega r$

$\therefore a = \frac{d(\omega r)}{dt} = r\frac{d\omega}{dt}$

$\Rightarrow a = r\alpha$

∴ Linear acceleration

= Radius × Angular acceleration

33. Define Work.

Answer:

Work: Work is defined as the product of displacement and the component of the force along the displacement. Work is a scalar quantity. The SI unit of work is Joule(J).

Mathematically, Work $W = \vec{F}.\vec{S} = FS\cos\theta$

34. Define the term Power. Is it a scalar or vector quantity? Give its dimension and unit in SI system.

Answer:

Power: The rate of work i.e. the amount of work done in unit time is called power.

Power$=\frac{Work}{Time}$ i.e. $P = \frac{W}{t}$

Power has magnitude only and no direction so it is a scalar quantity.

The SI unit of power is Joule/Second or Watt

Dimension of Power = $[ML^2T^{-3}]$

35. What is energy? Write SI unit of Energy.

Answer:

Energy: It is the capacity of a physical system to perform work. Energy exists in several forms such as heat energy, mechanical energy, light energy, sound energy or other forms.

The SI unit of energy is Joule.

36. 6) Write the mathematical expression of potential and kinetic energy.

Answer:

Potential Energy $E_p = mgh$

Where,

m = Mass of body

h = Height of body from ground level

g = Acceleration due to gravity

Kinetic Energy $E_k = \frac{1}{2}mv^2$

Where,

m = Mass of body

v = Velocity of the body

37. An apple of mass $0.1\ kg$ is dropped from $10\ m$ height. Find out the value of the kinetic energy when it hits the ground.

Solution:

By the principle of conservation of energy, the potential energy of the body at height h is converted into kinetic energy before it strikes the ground. Thus, $K_E = mgh$

$= 0.1 \times 9.8 \times 10$

$= 9.8$ Joule

∴ The value of kinetic energy is 9.8 Joule. (Ans.)

38. A man weighing $60\ kg$ climbs up a staircase carrying a load of $30\ kg$ on his head. The staircase has 20 steps each of height $0.3\ m$. If he takes $10\ s$ to climb, find his power.

Solution:

Here, Total mass $m = 60 + 30 = 90\ kg$

Total height lifted $h = 20 \times 0.3 = 6\ m$

∴ Work done = Potential energy gained

$= mgh$

$= 90 \times 9.8 \times 6 = 5292$ Joule

Time taken $t = 10\ s$

∴ Power $= \frac{W}{t} = \frac{5292}{10} = 529.2\ W$ (Ans.)

{Prepared by NatiTute YouTube Channel}

39. The power of a water pump is $2\ kW$. How much time will it take to lift a mass of $200\ kg$ of water to a height of $40\ m$?

Solution:

Given, mass of water $m = 200\ kg$

height lifted $h = 40\ m$

Work done = Potential energy

$= mgh$

$= 200 \times 9.8 \times 40$

$= 78400$ J

Given, power of pump $= 2\ kW = 2000\ W$

Now, Power $P = \frac{W}{t}$, $\Rightarrow t = \frac{W}{P} = \frac{78400}{2000} = 39.2\ s$

∴ Time required to lift water is 39.2 second.

{Prepared by NatiTute YouTube Channel}

40. A body moving with uniform acceleration covers 20m in 4th second and 30m in 8th second find-

i) the initial velocity

ii) the acceleration of the body

Answer: The formula for the distance traveled in the n^{th} second by a body moving with uniform acceleration is given by: $S_n = u + \frac{a}{2}(2n - 1)$

Applying given two condition in the equation above we get, $20 = u + \frac{a}{2}(2 \times 4 - 1)$

$\Rightarrow 40 = 2u + 7a$, $\Rightarrow 2u + 7a = 40$ .... (1)

And, $30 = u + \frac{a}{2}(2 \times 8 - 1)$

$\Rightarrow 60 = 2u + 15a$, $\Rightarrow 2u + 15a = 60$ .... (2)

Substituting (1) from (2)-

$15a - 7a = 60 - 40$

$\Rightarrow 8a = 20, \Rightarrow a = \frac{20}{8} = 2.5\ m/s^2$

Putting $a = 2.5$ in eq. (1)

$2u + 7 \times 2.5 = 40$

$\Rightarrow 2u = 40 - 17.5 = 22.5$

$\Rightarrow u = \frac{22.5}{2} = 11.25\ m/s$

∴ i) The initial velocity = $11.25\ m/s$ (Ans)

ii) The acceleration of the body = $2.5\ m/s^2$ (Ans)